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Chapter 19

GELFAND-P000954

16/II-81. Gelfand Sem.



Original source details: GELFAND-P000954

Diagram and typed description. A narrow hatched bar underlines the date. II has a stroke above and below it.

16/II-81.

Khovanskii. On Newton polyhedra.

[At the right: (reverse side of the prev. talk by Mc Pherson.)]



Original source details: GELFAND-P000954

Diagram and typed description. The note identifies this as the reverse side of the preceding talk.

$n = 2$ for simplicity.

refs.: 1) Mumford et al. Lect. Notes — toric embeddings.

2) D. Bernstein, FA.

1) Kempf, Knudsen, Mumford, Saint-Donat, Toroidal Embeddings,

Lect. Notes Math., 1972, $N = 339$.



Original source details: GELFAND-P000954

Diagram and typed description. The manuscript citation gives the year 1972 and number 339.

GELFAND-P000955

and also a survey by Danilov in UMN.

Articles by Khovanskii: FA, 1977, 11:4

1978, 12:1

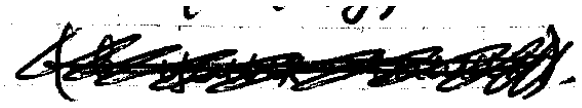
$$\{(x, y) : P(x, y) = 0\} \tag{1}$$

what does this curve look like? (handles, etc.)

How many solutions does the system have

$$\begin{cases} P(x, y) = 0 \\ Q(x, y) = 0 \end{cases} \tag{2}$$

[crossed out: (line unresolved).]



Original source details: GELFAND-P000955

Undeciphered passage. The parenthetical line is heavily crossed out.

$$y^2 = P_n(x) \tag{3}$$

has $\sim n/2$ handles (more precisely, $\left[\frac{n-1}{2}\right]$ handles).

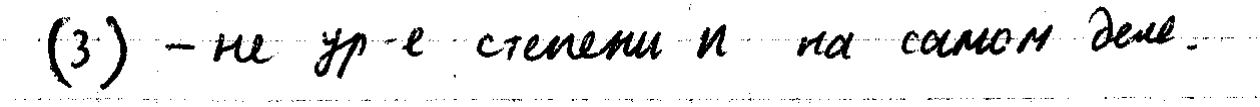
$$\begin{aligned} n = 3, 4 \quad & 1 \text{ handle}, \\ n = 5, 6 \quad & 2 \text{ handles}. \end{aligned}$$

Condition: P_n has no multiple roots.

For a general eq. (1) there is also a recipe for computing the number of handles:

$$\frac{(n-1)(n-2)}{2} \qquad \text{for an eq. in general position.} \tag{4}$$

(3) — is not an eq. of degree n in fact.

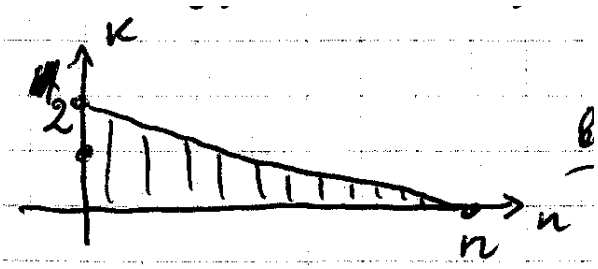


Original source details: GELFAND-P000955

Diagram and typed description. The sentence about the degree in (3) follows the manuscript wording.

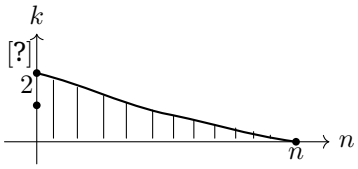
It all comes down to the Newton polyh..

$$P(x, y) = \sum a_{n,k} x^n y^k. \tag{5}$$



Original source details: GELFAND-P000955

Diagram and typed description. Hatched triangle, axes k, n, and marked points. The label above 2 is crossed out.

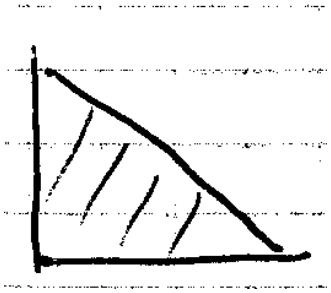




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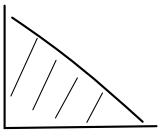
Undeciphered passage. The heavily crossed-out label above 2 is unresolved.

Convex hull of those (n, k), for which $a_{nk} \neq 0$.



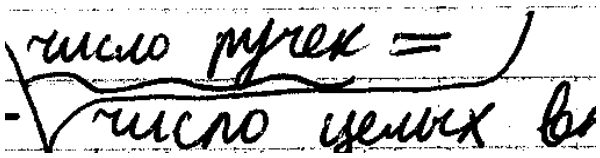
Original source details: GELFAND-P000955

Diagram and typed description. The second triangle, with four interior strokes.



— gen. case — simplex.

Answer for (1) — [inserted above the line: number of handles =] number of interior lattice points in the Newton polyh.



Original source details: GELFAND-P000955

Diagram and typed description. "number of handles" and the equals sign are inserted above the answer.

(for coeffs. $a_{n,k}$ in general position)

This is a theor. of Baker (that very one), rediscovered by Khovanskii in 1976.

(Baker taught Hodge, Hodge taught Atiyah)

In $\mathbb{C}^2$ some points of the curve (1) are hidden. We want to bring them out by certain substitutions.

Consid. the curve $P(x, y) = 0$ away from the coordinate cross.

x, y — meromorphic fns. on a Riem. surface.

Sometimes they become 0 or ∞. This gives a divisor:

at $\forall$ of its points there is a vector (n, m)

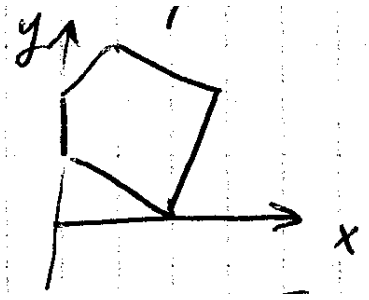
(order of the zero of $x = n$, order of the zero of $y = m$ at the given point)

GELFAND-P000956

[source number: 6]

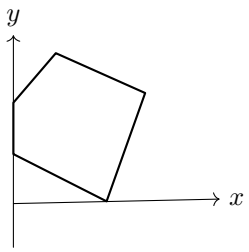
Points of order $(1, 0)$ or $(k, 0)$ are visible in the picture: they are the points of inters. with the axis $x = 0$.

We can transl. the polygon by multiplication by x^k and y^k.



Original source details: GELFAND-P000956

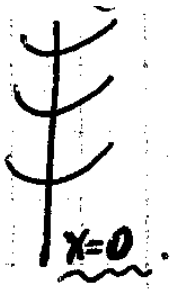
Diagram and typed description. The pentagon meets the y-axis along an edge and the x-axis at a vertex.



Translate so that it touches the axis $x = 0$.

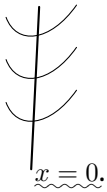
$$P(0, y) = 0 \tag{6}$$

Number of sols. $=$ length of the segment of intersection of NP with the axis $\{x = 0\}$.
(if there are no multiple roots



Original source details: GELFAND-P000956

Diagram and typed description. Three branches cross the vertical line labelled $x = 0$.



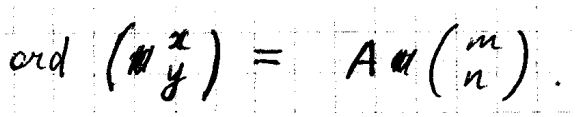
nonmultiple $\Rightarrow$ the implicit fn. thm. applies
$\Rightarrow$ there are no points of the form $(k, 0)$, $k \geqslant 2$.
Make a transf.

$$\begin{cases} x = u^{a_{11}} v^{a_{12}}, \\ y = u^{a_{21}} v^{a_{22}}, \end{cases} \qquad a_{ij} \in \mathbb{Z} \tag{7}$$

$$\det \begin{Vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{Vmatrix} = 1, \qquad A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}. \tag{8}$$

the inverse is integer-valued $\Rightarrow$ there is an inverse transf. (also rational)
If $u \sim t^m$, $v \sim t^n$, then after the transformation

$$\text{ord} \begin{pmatrix} x \\ y \end{pmatrix} = A \text{ [crossed out]} \begin{pmatrix} m \\ n \end{pmatrix}. \tag{9}$$



Original source details: GELFAND-P000956

Uncertain reading. The signs left of the column x, y and after A are heavily crossed out. Both resemble u, but the exact reading is unresolved.

We want a transf. such that at the pt. $\binom{n}{m}$ there lands $\binom{1}{0}$.
Let the vector (n, m) be primitive ($\sim$ on the segment $[0, (m, n)]$ there are no lattice points).



Original source details: GELFAND-P000956

Uncertain reading. The first character of the pair is heavily retraced and tentatively read as n. The next entry has the order (m, n).

Moral: ∃ a matrix A.

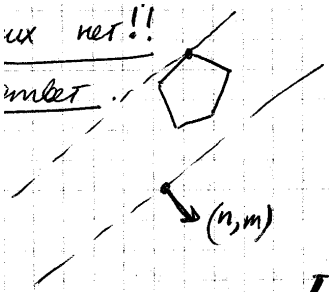
If [crossed out: ~~it~~ (?); inserted above the line: (n, m)] is reducible, then this same transf. shows that there are no such divisors!!



Original source details: GELFAND-P000956

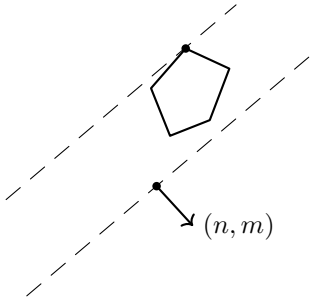
Uncertain reading. Above the crossed-out word, tentatively "it", is the insertion (n, m).

answer.



Original source details: GELFAND-P000956

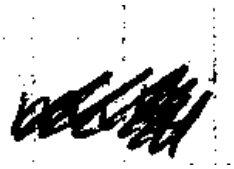
Diagram and typed description. One of the parallel lines meets a vertex of the pentagon. An arrow labelled (n, m) starts at the marked point on the other line.



move it until it touches

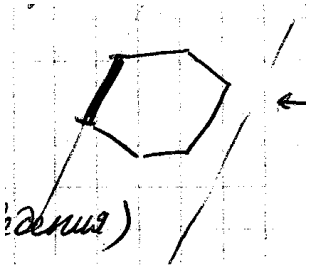
[crossed out: ~~if~~ (?)] We hit a point —

— then there are none of this kind.



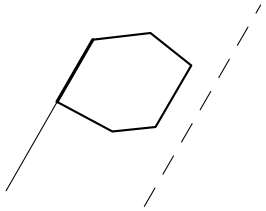
Original source details: GELFAND-P000956

Uncertain reading. The crossed-out word is tentatively read as "if".

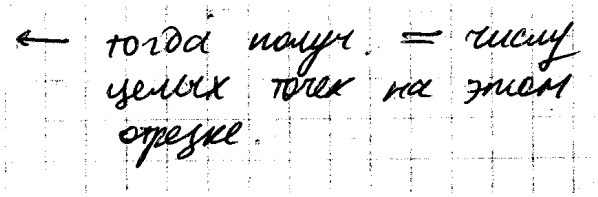


Original source details: GELFAND-P000956

Diagram and typed description. The left edge of the hexagon is emphasized on a sloping line. A parallel line is to the right.



← then we get = the number of lattice points on this segment.



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Diagram and typed description. The last line gives "the number of lattice points on this segment".

(take the min of the scalar product)

GELFAND-P000957

Total number of punctures = the number of lattice points on the boundary.

Consider now the system

$$\begin{cases} P(x,y)=0 \\ Q(x,y)=0 \end{cases} \tag{10}$$

and seek the number of sols. in $(\mathbb{C}\setminus 0)^2$

([crossed out: unresolved] $x\neq 0,\infty$; $y\neq 0,\infty$).



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Uncertain reading. A short entry before the condition on x is obscured by crossing-out.

Consid. the fn. Q on the curve $P(x,y)=0$.

Its divisor $=0$,



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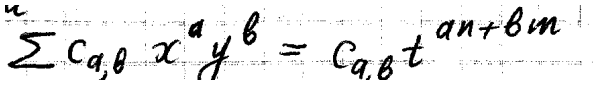
Diagram and typed description. The manuscript reads “Its divisor $=0$”.

Look at the order at the omitted pts.!!

(n,m) $\quad x\sim t^n,\ y\sim t^m$.

Then

$$\sum C_{a,b}x^a y^b = C_{a,b}t^{an+bm} \tag{11}$$



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Diagram and typed description. In (11) the summation sign stands to the left of the equality.

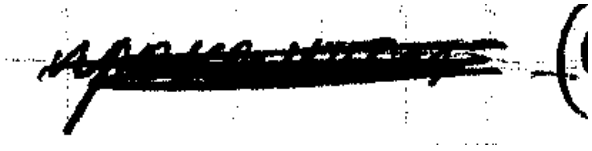
answer:

$$\underline{\min_{a,b}\big((n,m)\cdot(a,b)\big)} \quad \text{— a purely geom. answer.} \tag{12}$$

Punctured points — on the cross or at ∞

$\Rightarrow$ poles only at the punctured points.

$$\left[\sum_{m,n}(12)\right] = \text{the number of sols. of (10) in [crossed out: ~~the space~~ (?)] } (\mathbb{C}\setminus 0)^2. \tag{13}$$



Original source details: GELFAND-P000957

Uncertain reading. The crossed-out word before $(\mathbb{C}\setminus 0)^2$ is tentatively read as “the space”.

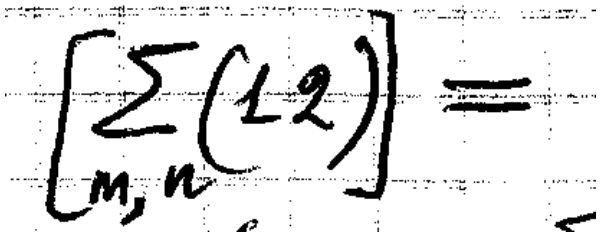


Diagram and typed description. There is no minus sign before the summation in (13).

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(for gen. pos. — otherwise cancellation may occur).

There are 2 polygons!! What do we get?

Answer — mixed volume!!

Add all the vects. (n, m) — we get a vector of length $=$ the length of the side.



Diagram and typed description. Three successive arrows point down and to the right.



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A vector $\perp$ to the side and equal to it in length.

Suppose the 2 polygons coincide (i.e. the same nonzero coefficients! of P and Q).

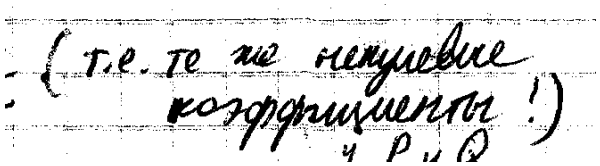


Diagram and typed description. The parenthesis reads “the same nonzero coefficients!”

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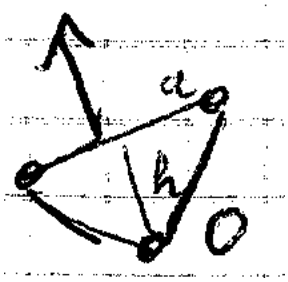
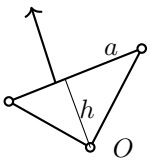


Diagram and typed description. Triangle with side a, height h, vertex O, and an outward arrow.



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max for the triangle among all scalar prods. $=$

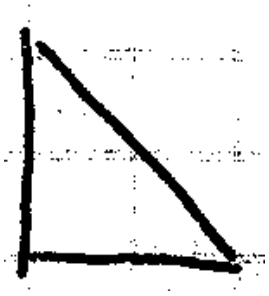


Diagram and typed description. The first word reads Latin max.

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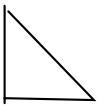
$= ah = 2S.$

(fix O)



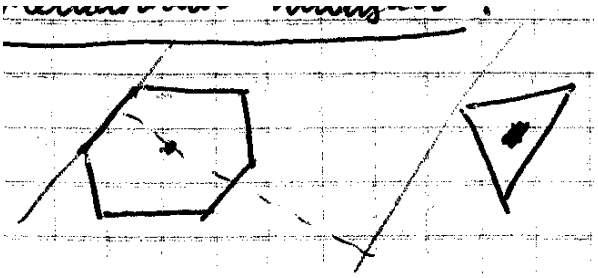
Original source details: GELFAND-P000957

Diagram and typed description. Small unlabelled right triangle.



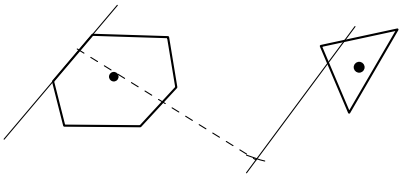
$\triangle$-gon $\Rightarrow$ we get n^2 (Bézout's theor.).

Mixed area.

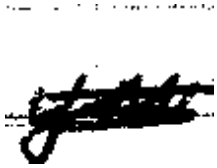


Original source details: GELFAND-P000957

Diagram and typed description. Hexagon and triangle with marked points. The dashed line and short strokes at the right-hand line are drawn approximately.



sides [crossed out: ~~mult.~~ (?)] of one mult. by the heights of the other.



Original source details: GELFAND-P000957

Uncertain reading. The crossed-out word after "sides" is tentatively read as "mult."

A common point is marked O.

$\forall$ side is moved until it supports the 2nd — this is the height.

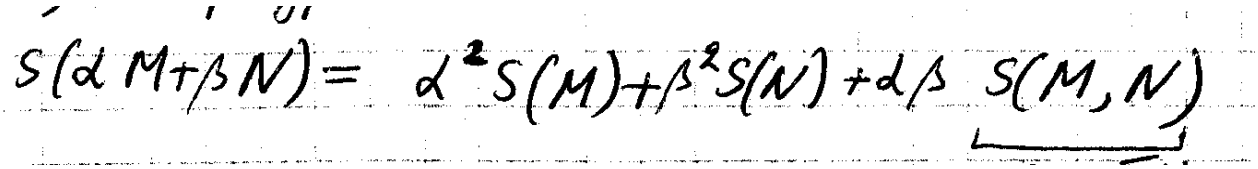
GELFAND-P000958

[source number: 7]

General defn.

M, N — figures.

$$S(\alpha M + \beta N) = \alpha^2 S(M) + \beta^2 S(N) + \alpha\beta \underbrace{S(M,N)}_{\text{mixed volume.}}$$



Original source details: GELFAND-P000958

Diagram and typed description. The first equality has $S(M,N)$ as the coefficient of $\alpha\beta$. The manuscript then uses V.

In n-dimensional space — mixed volume n figures.
$V(\Delta_1, \ldots, \Delta_n)$ is def. by the properties:

1) $V(\Delta, \ldots, \Delta) = V(\Delta)$ — volume.
2) $V(\Delta_1 + \Delta_1', \Delta_2, \ldots) = V(\Delta_1, \Delta_2, \ldots) + V(\Delta_1', \Delta_2, \ldots)$

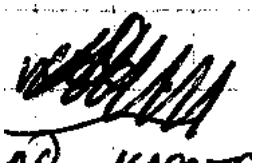
Such a fn. [crossed out: entry unresolved] exists and is unique.



Original source details: GELFAND-P000958

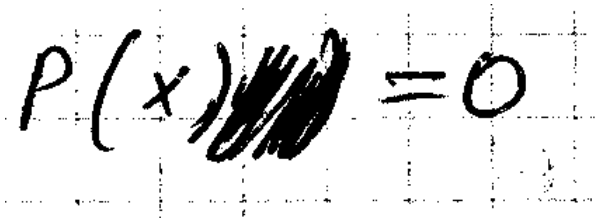
Uncertain reading. A short entry after "fn." is crossed out.

(14) Thm. (D. Bernstein)
Number of sols. (in $(\mathbb{C} \setminus 0)^2$) of a system in gen. pos. of form (2) $=$
$=$ mixed volume of their Newton polyh..
(15) Thm.
Euler char. [crossed out: ~~of the curve~~ (?)] of the hypersurf., $P(x)$ [crossed out: $, y)$ (?)] $= 0$



Original source details: GELFAND-P000958

Uncertain reading. The crossed-out word before "hypersurf." is tentatively read as "of the curve".

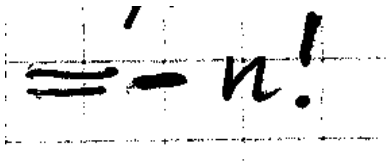


Original source details: GELFAND-P000958

Uncertain reading. A heavily crossed-out entry follows $P(x)$, possibly $, y)$.

(with the coord. cross removed)

$$= -n!(\text{volume of the Newton polyh.})$$



Original source details: GELFAND-P000958

Diagram and typed description. A minus sign is written before $n!$.

Corollary (geom.)

$$2 - 2B - \Gamma = -2S.$$

B — number of interior lattice pts..

Γ — —"— boundary —"—

$$B + \tfrac{1}{2}\Gamma = S + 1. \tag{16}$$

Relation of mixed volume to the isoperimetric inequality.

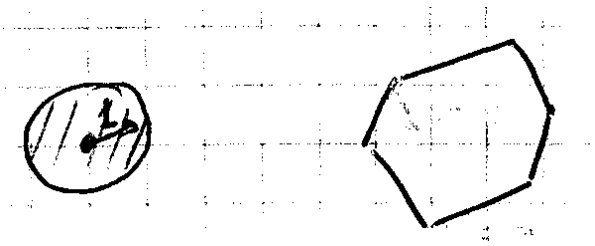
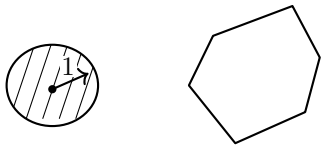


Diagram and typed description. Hatched circle with radius 1 and a hexagon.



Original source details: GELFAND-P000958

$$S(\Delta_1, \Delta_2) = S(\Delta_1 + \Delta_2) - S(\Delta_1) - S(\Delta_2). \tag{17}$$

O — centre of the circle.

Mixed area of the polygon and the circle = the perimeter of the polygon.

Inequality:

$$V^2(\Delta_1, \Delta_2) \geqslant V(\Delta_1)V(\Delta_2). \tag{18}$$

Attained on the circle.

Theorem for any alg. surface.

Γ_1, Γ_2 — two curves on an algebr. surface.

$$(\Gamma_1, \Gamma_2)^2 \geqslant (\Gamma_1, \Gamma_1) \cdot (\Gamma_2, \Gamma_2). \tag{19}$$

[label by (Γ_1, Γ_2): inters. index]

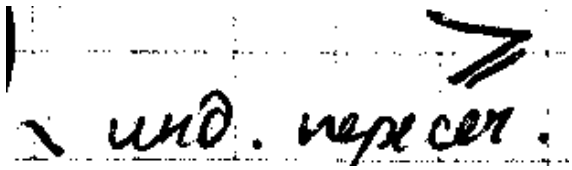
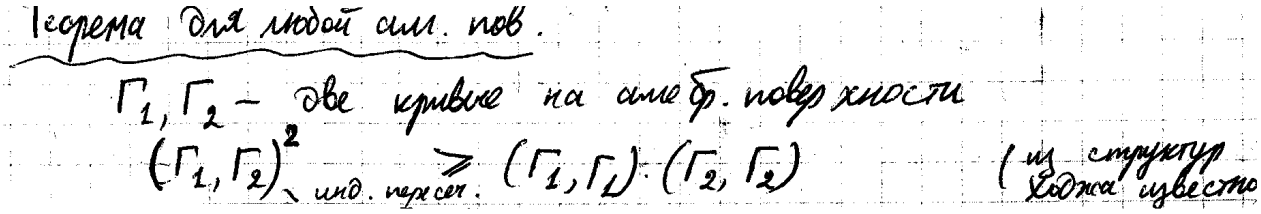


Diagram and typed description. A short stroke leads from the term on the left to "inters. index".

Original source details: GELFAND-P000958

(known from Hodge structures)



Original source details: GELFAND-P000958

Diagram and typed description. On this frame, (19) is accompanied by the two curves on an algebraic surface and the note on Hodge structures.

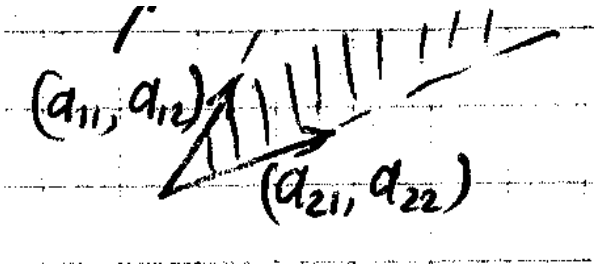
GELFAND-P000959

We construct a special compactif. of $(\mathbb{C} \setminus 0)^2$.

There is a set $P(x, y) = 0$.

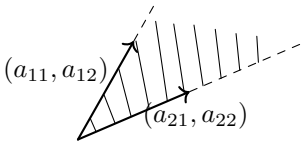
We want to describe its charts so that the transitions are only monomial and yet it would be possible to consider everything,

what compact complex manifolds there are which are covered by a system of charts, the transition being given by unimod. matrices.

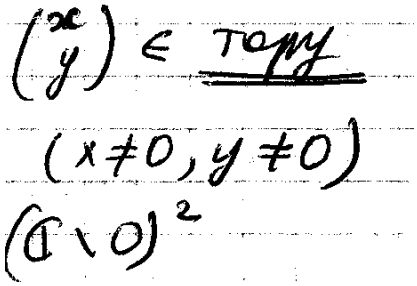


Original source details: GELFAND-P000959

Diagram and typed description. Hatched cone with two labelled rays. The rays continue in broken strokes beyond the arrowheads.



$$\begin{pmatrix} x \\ y \end{pmatrix} \in \underline{\underline{\text{the torus}}}$$
$$(x \neq 0, y \neq 0)$$
$$(\mathbb{C} \setminus 0)^2$$



Original source details: GELFAND-P000959

Diagram and typed description. The word “ropy” (“the torus”) is underlined twice.

[under the figure on the left: crossed out ~~it~~ (?)]

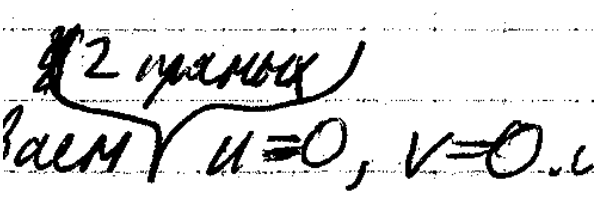


Original source details: GELFAND-P000959

Uncertain reading. The crossed-out entry resembles “он” (“it”); the reading is unresolved.

For now it is noncompact.

With each transf. (7) we glue on $u = 0$, $v = 0$ [above the entry in a brace: “2 lines”] and draw a cone.



Original source details: GELFAND-P000959

Uncertain reading. At the start of the brace, left of “2 lines”, there is a dark stroke. It is unclear whether a separate entry was crossed out here.

If they intersect, the construction is impossible.

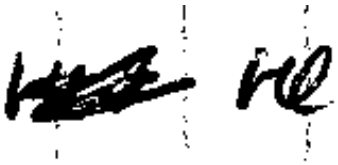
It is possible if the cones cover all of $\mathbb{R}^2$.

GELFAND-P000960

[source number: 12]

23/II–81. Gelfand seminar.

I.M.G. Dixmier found an example of 2 oper. A_1 and A_2 of orders 4 and 6, [crossed out: ~~not~~ (?)] that are not functions of one oper. A_0 of 2nd order; A_1, A_2 — with polynomial coeff. (and leading coeff. 1), such that $[A_1, A_2] = 0$.



Original source details: GELFAND-P000960

Uncertain reading. The crossed-out entry before “are not” tentatively reads “не” (“not”).

$$A_2 = \sqrt{\underbrace{\left(\frac{d^4}{dx^4} + f_0(x)\frac{d^3}{dx^3} + \dots\right)^3}_{A_1^3} + C}.$$

Grinevich found many such examples via Tyurin parameters.

Cherednik I.

On an event that brings soliton theory to a close.

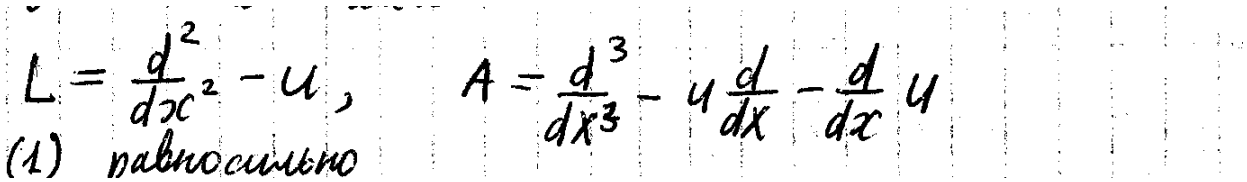
Stage I. Inverse problem method.

$$u_t = 6uu_x + u_{xxx}. \tag{1}$$

$$L = \frac{d^2}{dx^2} - u, \qquad A = \frac{d^3}{dx^3} - u\frac{d}{dx} - \frac{d}{dx}u. \tag{2}$$

Then (1) is equivalent to

$$[L, A] = L_t. \tag{3}$$



Original source details: GELFAND-P000960

Diagram and typed description. In (2), a minus sign precedes each of the two terms with u. No numerical coefficient precedes $\frac{d^3}{dx^3}$.

$$L \rightsquigarrow \text{scattering matr. } S(\lambda) \qquad (2\times 2 \text{ matrix}). \tag{4}$$

$S(\lambda)$ changes in a simple way as one varies t.

Stage II. (Zakharov et al.)

$$\left[\underbrace{\frac{d}{dx}-U}_{L}, \frac{d}{dt}-V\right]=0 \qquad \text{(zero-curvature representation)}. \tag{5}$$

U, V — rat. functions of λ.

For example,

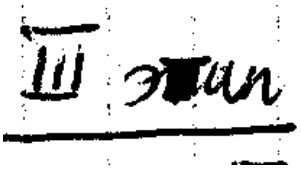
$$\begin{cases} U = U_0 + U_1\lambda + U_2\lambda^2, \\ V = V_0 + \lambda V_1 \end{cases} \qquad — \tag{6}$$

and one obtains a system of eqs.

$L \rightsquigarrow S$ and so on.

The Hamiltonian character was observed.

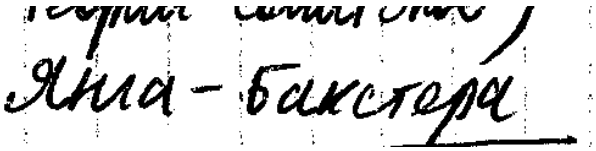
Stage III (the twilight of soliton theory)



Original source details: GELFAND-P000960

Uncertain reading. One stroke in the word “этап” (“stage”) is heavily retraced. The underlying entry is unresolved.

Yang–Baxter identities.

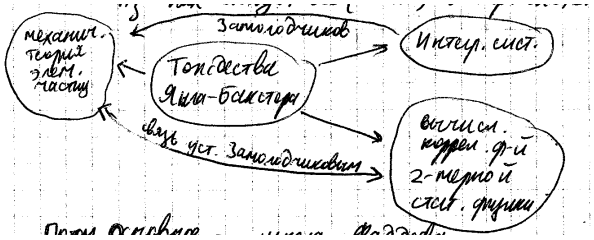


Original source details: GELFAND-P000960

Diagram and typed description. The manuscript spells the name “Яна-Бакстера”, without the Russian letter “г”.

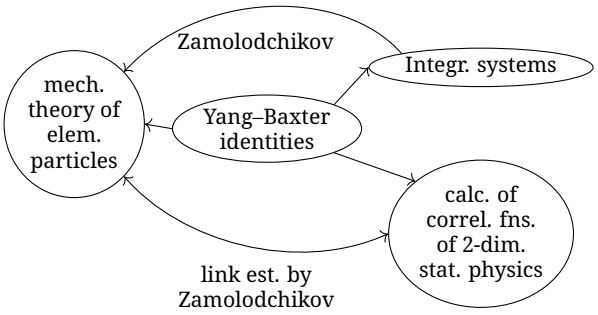
Algebr. object $\rightsquigarrow$ everything, with minor nuances.

From them one obtains all (almost) integr. systems.

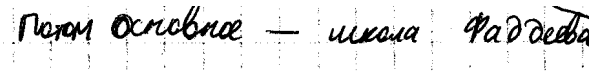


Original source details: GELFAND-P000960

Diagram and typed description. Three arrows leave the middle oval. The upper arc points left, and the lower arc has arrowheads at both ends.



Then (?) the main thing is the Faddeev school.



Uncertain reading. The first word of the last line looks closer to “Потом” (“Then”); the strokes remain indistinct.

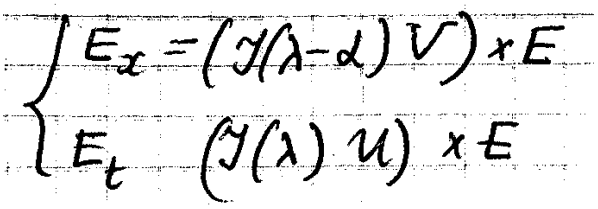
Original source details: GELFAND-P000960

GELFAND-P000961

Landau–Lifshitz eqs. and ... chir. fields

$$\left[\frac{d}{dt} - U, \frac{d}{dx} - V\right] = 0, \qquad \lambda \in \text{ell. curve!!} \tag{7}$$

$$\begin{cases} E_x = (J(\lambda - \alpha)V) \times E, \\ E_t \quad (J(\lambda)U) \times E \end{cases} \tag{8}$$



Uncertain reading. In the second line of (8) no sign is visible between E_t and the parenthesis.

Original source details: GELFAND-P000961

E, V, U — 3-vectors.

shift on the ell. curve

$$J = \text{diag}(\cdot, \cdot, \cdot).$$

hence

$$\begin{cases} U_x = (JV) \times U, \\ U_t = (JU) \times V \end{cases} \qquad \text{(asymmetric principal chiral field).} \tag{9}$$

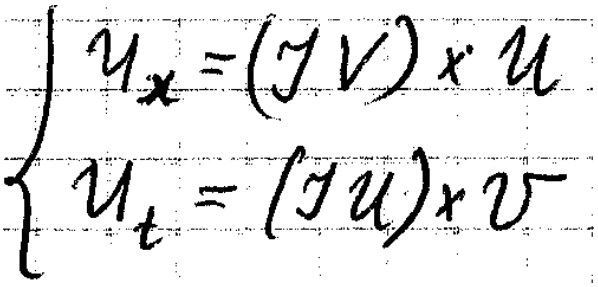


Diagram and typed description. The second line of (9) begins with U_t.

Original source details: GELFAND-P000961

J — const. diagon. 3-matrix.

$$J = J(\alpha).$$

The matrix J — arbitrary!!

(Attainable by choice of α and the ell. curve).

It is integrated using the ell. curve.

Landau–Lifshitz eq.:

$$U_t = U \times U_{xx} + JU \times U. \tag{10}$$

The Y.–B. identity at once gives the entire Hamiltonian formalism together with the brackets of the scattering data.

Khovanskii A.

We want:

$\overset{\swarrow \text{manifold}}{M^n} \supset (\mathbb{C} \setminus 0)^n$ we will construct a special compactif. of the torus.

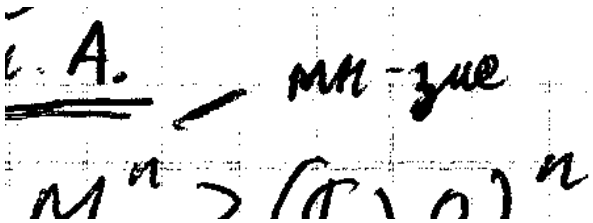


Diagram and typed description. The note “мн-зие” (“manifold”) above M^n is linked to it by a slanting stroke.

Original source details: GELFAND-P000961

and the complement is an analyt. set [crossed out: mark unresolved] of smaller dimension.



Uncertain reading. A short crossed-out mark is visible before “smaller”; its reading is unresolved.

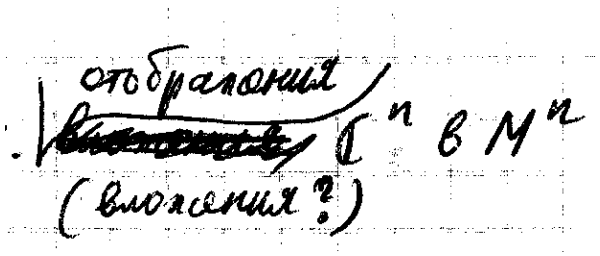
Original source details: GELFAND-P000961

Modular charts:

$$\begin{cases} x_1 = u_1^{a_{11}} \dots u_n^{a_{1n}} \\ \dots\dots\dots\dots \\ x_n = u_1^{a_{n1}} \dots u_n^{a_{nn}} \end{cases} \tag{1}$$

and extend to analyt. [crossed out: ~~embeddings~~ (?); inserted above: maps] $\mathbb{C}^n$ into M^n.

[beneath the revised word:] (embeddings?)



Uncertain reading. “Maps” is written above the cancellation. Below it one reads “(embeddings?)”; the cancelled word may be the same.

Original source details: GELFAND-P000961

One-param. subgroup (of the torus):

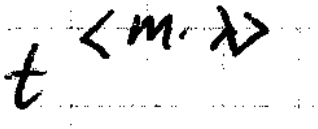
$$\begin{cases} x_1 = t^{\lambda_1} \\ \quad \vdots \\ x_n = t^{\lambda_n} \end{cases} \tag{2}$$

GELFAND-P000962

[source number: 13]

$$\underline{\text{Monomial}} \quad x_1^{m_1} \dots x_n^{m_n} \tag{3}$$

behaves as $t^{\langle m,\lambda \rangle}$ on subgroup (2).



Uncertain reading. The short stroke between m and λ is rendered as a comma. It is close to a dot.

Original source details: GELFAND-P000962

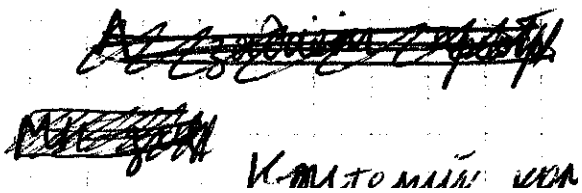
For $\forall$ chart we draw a cone in the space of one-parameter subgroups.
They are arrang. in a certain way.

$$\begin{pmatrix} a_{11} \\ \vdots \\ a_{n1} \end{pmatrix}, \dots, \begin{pmatrix} a_{1n} \\ \vdots \\ a_{nn} \end{pmatrix}$$ generate a conv. cone.

Intersect in their interiors $\Rightarrow$ they coincide. (condition for existence of the manifold)
Example:

$$\begin{cases} x_1 = u_1 u_2^{-1} \\ x_2 = u_2 \end{cases} \qquad \begin{cases} x_1 x_2 = u_1 \\ x_2 = u_2 \end{cases}$$

This is a σ-process.



Undeciphered passage. Two entries are crossed out, one before “If the faces” and one at the start of the next line. The words under the dense strokes are unresolved.

Original source details: GELFAND-P000962

If the faces intersect in their interiors, they likewise coincide.

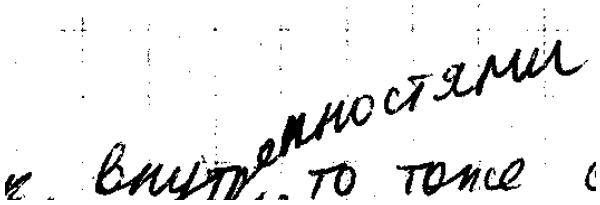
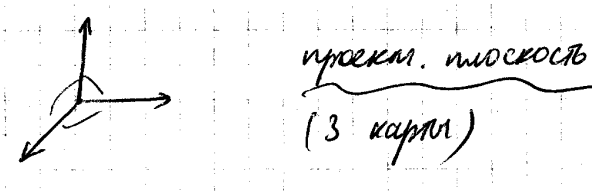


Diagram and typed description. The Russian ending “енностями”, completing “interiors”, is added diagonally after the abbreviation.

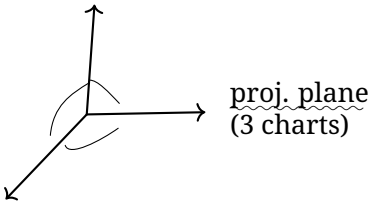
Original source details: GELFAND-P000962

Compactness criterion: nec. and suff. that the cones cover the entire space.



Original source details: GELFAND-P000962

Diagram and typed description. Three directed rays and three arcs without arrowheads.



The compactif. is specified by a subdivision of the integer space up to an integer automorphism.

Kazarnovskii.

Problem: Generalize the theorem on zeros of polynomials to functions of a more general form.

$$f = \sum_{\lambda \in \Lambda} C_\lambda \exp\langle z, \lambda \rangle \tag{1}$$

regarded as functions in $\mathbb{C}^n$; $\lambda \in \Lambda \subset \mathbb{C}^n$, $z \in \mathbb{C}^n$

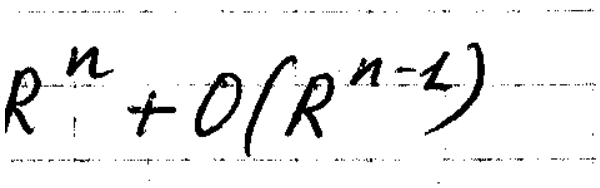
$$\Gamma = \mathrm{conv}(\Lambda) \qquad \text{— Newton polyh.}$$

Consider the system

$$F : \{f_1 = \ldots = f_n = 0\}.$$

The number of zeros in the ball of rad. R is denoted $N_F(R)$

$$N_F(R) = cV(\Gamma_1, \ldots, \Gamma_n)R^n + O(R^{n-1}). \tag{2}$$



Original source details: GELFAND-P000962

Diagram and typed description. The exponent in the remainder term of (2) is written $n - 1$.

$V(\Gamma_1, \ldots, \Gamma_n)$ — mixed volume if all the polyh. lie in the real parts of $\mathbb{C}^n$

1 variable $\Rightarrow$ the perimeter of a polygon appears

c — const, consisting of powers of 2 and π.

GELFAND-P000963

What is V in the general case?

$$\Gamma \subset \mathbb{C}^n,$$

$$V(\Gamma) = \sum_{\dim \Delta = n} \underset{\substack{\downarrow \\ \text{volume of face}}}{V(\Delta)} \quad \Omega(\Delta) S(\Delta).$$

Faces of middle dim. (n [mark unresolved])



Original source details: GELFAND-P000963

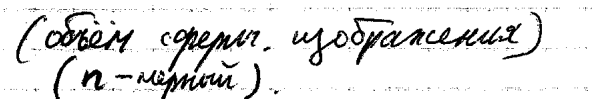
Uncertain reading. A heavy stroke follows n inside the parentheses. An exclamation mark is possible, but no separate dot is visible.

$\Omega(\Delta)$ — [crossed out: mark unresolved] ext. angle (volume of the spher. image) [under "volume of the spher. image":] (n-dimensional).



Original source details: GELFAND-P000963

Uncertain reading. A short entry before "ext." is crossed out. The strokes obscure the reading.

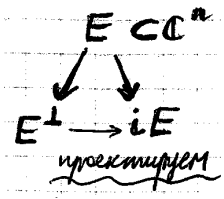


Original source details: GELFAND-P000963

Diagram and typed description. "(n-dimensional)" is written below the parenthetical phrase about the volume of the spherical image.

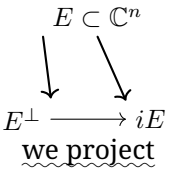
$S(\Delta)$ is def. as follows

Consider the real Grassmannian $G_{2n,n}$; more precisely, real n-dim. subsp. in $\mathbb{C}^n$.



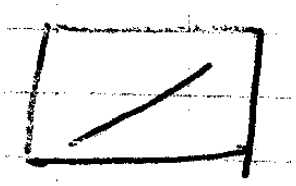
Original source details: GELFAND-P000963

Diagram and typed description. Two arrows leave $E \subset \mathbb{C}^n$. The horizontal arrow goes from $E^\perp$ to iE.



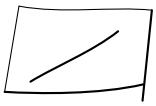
area multiplied by the number denoted $S(\Delta)$.

$V(\Gamma)$ extends from polyhedra to arbitrary bodies $M \subset \mathbb{C}^n$.



Original source details: GELFAND-P000963

Diagram and typed description. Unlabelled quadrilateral with a slanting stroke inside. Its right side extends below the bottom edge.



T_cM — complex tang. plane to ∂M.

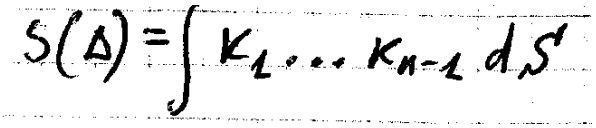
B — 2nd quadr. form, def. on T_cM

The quadr. form $\dfrac{B(ix)+B(x)}{2}$ def. a Hermitian form

(Levi form)

$\kappa_1, \ldots, \kappa_{n-1}$ — eigenvalues.

$$S(\Delta) = \int \kappa_1 \ldots \kappa_{n-1}\, dS$$



Original source details: GELFAND-P000963

Diagram and typed description. The left side of the last formula is written $S(\Delta)$.

Another version: via the support fn.

GELFAND-P000964

2.3.81. Gelfand seminar.

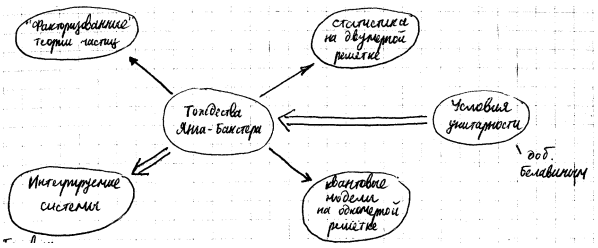
I.M.G. Problem. $\int_x^\infty e^{-t^2}\,dt$ expand in an asympt. series which diverges but is Borel summable.

Problem. Relate Kazarnovskii's result to the growth indicatrix.

GELFAND-P000965

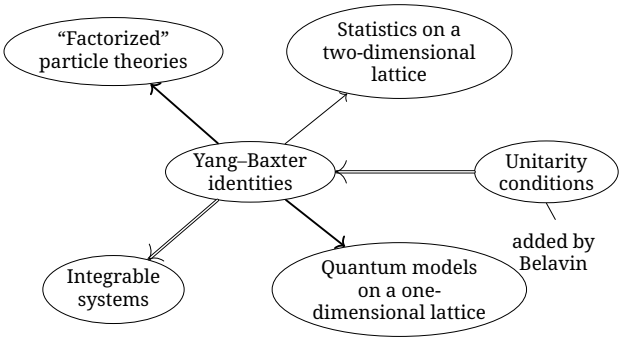
[source number: 19]

Cherednik I.

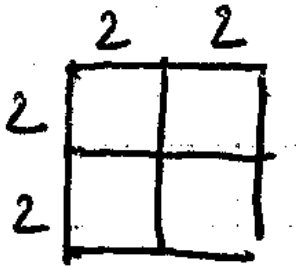


Original source details: GELFAND-P000965

Diagram and typed description. Six ovals. The double arrow on the right points toward the identities, and the one at the lower left points away from them. The middle oval spells the name "Янa-Бакстера".

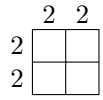


Belavin.



Original source details: GELFAND-P000965

Diagram and typed description. Four blocks, each with two rows and two columns.

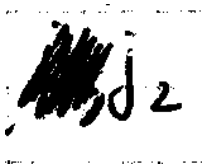
(1)  $= R^{j_1 j_2}_{i_1 i_2} \longrightarrow R^{1\,2}$

(abbrev. notation)

Gen. case: $n^2 \times n^2$.

i_1, j_1 index the blocks

i_2 [crossed out: entry unresolved], j_2 — entries within a block.



Original source details: GELFAND-P000965

Uncertain reading. The entry between i_2 and j_2 is obscured by strokes.

Interpr. as a tensor product.

$$R^{12} \xrightarrow{\Omega} (R^{12})^{-1} \qquad \underline{\text{unitarity transf.}} \tag{2}$$

$$R^{12} \xrightarrow{C} R^{\bar{1}2} \qquad \underline{\text{crossing}} \tag{3}$$

(Block transp. is crossing)
within the blocks it is unchanged

$$\Omega^2 = C^2 = 1 \tag{4}$$

$$(\Omega C)^n \neq 1 \tag{5}$$

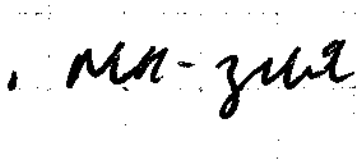
(6) [crossed out: entry unresolved] $R^{12} \longrightarrow (\Omega C) R^{12}$



Original source details: GELFAND-P000965

Uncertain reading. The first entry in (6) is crossed out. Its symbols cannot be made out reliably beneath the strokes.

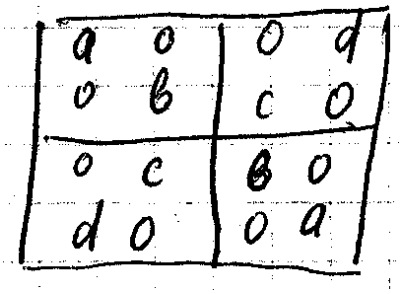
Invar. varieties — ? (for (6)).



Original source details: GELFAND-P000965

Diagram and typed description. The abbreviation after “Invar.” reads “мн-зия” (varieties).

Consider matr. of Baxter type



Original source details: GELFAND-P000965

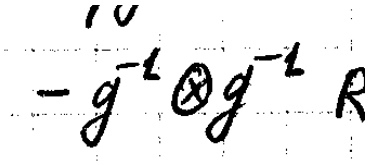
Diagram and typed description. Four matrix blocks. In row three, b is repeated on the diagonal.

(7)

a	0	0	d
0	b	c	0
0	c	b	0
d	0	0	a

(invar. with resp. to 2 groups Z_2 ⇓)

$$R^{12} \longrightarrow -g^{-1} \otimes g^{-1}\ R^{12}\ g \otimes g \tag{8}$$

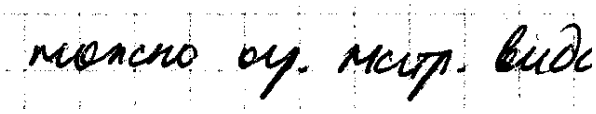


Original source details: GELFAND-P000965

Diagram and typed description. In (8) a separate minus sign precedes the first g^{-1}.

$$g_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad g_2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{9}$$

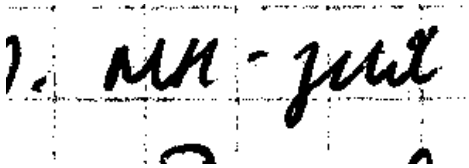
Transf. (8) together with $\Omega C \Rightarrow$ one can restrict [abbrev.] to matr. of form (7).



Original source details: GELFAND-P000965

Uncertain reading. The abbreviation before “matr.” resembles “or.”, tentatively expanded as “restrict to”.

Invar. varieties are as follows:



Original source details: GELFAND-P000965

Diagram and typed description. The abbreviation “мн-зия” (varieties) recurs before “are as follows”.

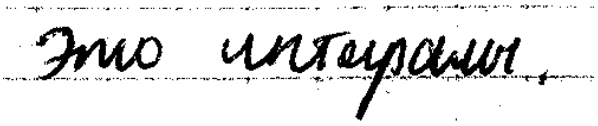
(with resp. to Ω and C simultaneously)

GELFAND-P000966

$$\frac{a^2 + b^2 - c^2 - d^2}{ab} = I_1 \tag{10a}$$

$$\frac{cd}{ab} = I_2 \tag{10б}$$

These are integrals.



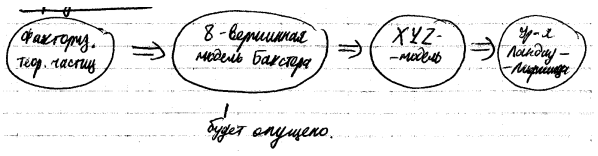
Original source details: GELFAND-P000966

Diagram and typed description. Under (10б) the manuscript reads “These are integrals”.

This is an ellipt. curve.

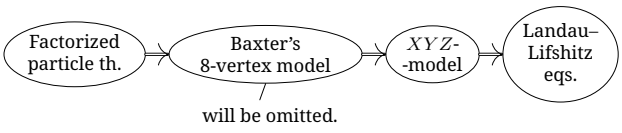
— · —

Cherednik I.



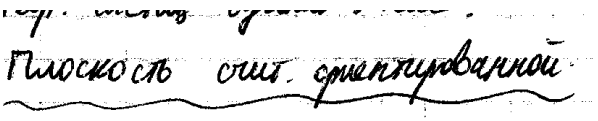
Original source details: GELFAND-P000966

Diagram and typed description. Four ovals with double arrows pointing right. Under the Baxter oval the note reads “will be omitted”.



A th. of particles of equal masses is constructed in 2-dim. space/time.

d types of particles The plane is taken as oriented



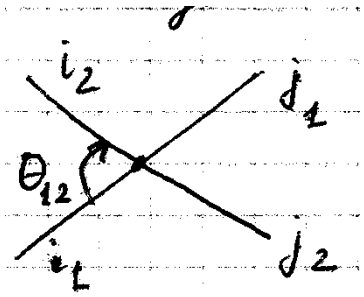
Original source details: GELFAND-P000966

Diagram and typed description. The final word of the wavy-underlined line reads “ориентированной” (“oriented”).

Momenta are conserved.

The only event is the intersect. of lines and their recoloring

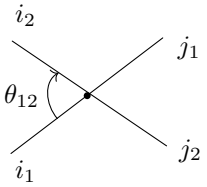
(elementary event)



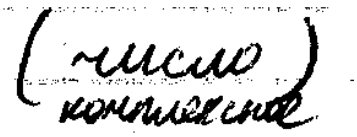
Original source details: GELFAND-P000966

Diagram and typed description. Two intersecting lines. The arc of angle θ_{12} on the left points upward.

(1)



(2) Assign an amplitude $S^{j_1 j_2}_{i_1 i_2}$ (number [inserted below: complex])



Original source details: GELFAND-P000966

Diagram and typed description. "Complex" is inserted below "number".

(3) $|S|^2$ — the probability of what took place

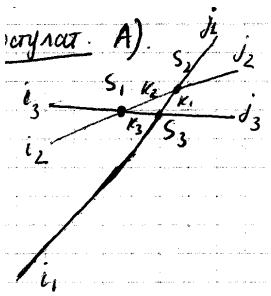
Supp. ($\Leftrightarrow$ relativism)

$$S^{j_1 j_2}_{i_1 i_2}(\theta_{12}), \tag{4}$$

θ_{12} — the usual Euclidean oriented angle

Nonrelativ. $\Leftrightarrow$ difference of velocities.

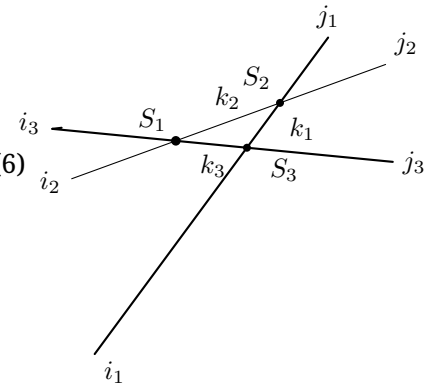
(5) Postulate. A).



Original source details: GELFAND-P000966

Diagram and typed description. Three pairwise intersections. The thick stroke at the left end of $i_3 j_3$ may be the start of the line or an incomplete arrowhead.

(6)



More complex event

It is assigned an amplitude S

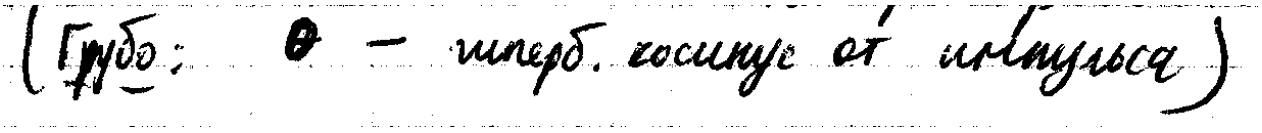
S_j — transition amplit. at the points indic. in the diagram

$$S = S_1 S_2 S_3 \tag{7}$$

B) The amplitude is independent of the location of events at a point in space,

i.e. it is unchanged by parallel translation of the lines.

(Roughly: θ — hyperb. cosine of the momentum) (we keep the colors of the outer ends)

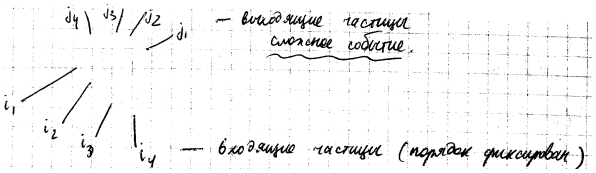


Original source details: GELFAND-P000966

Diagram and typed description. The lower entry reads “hyperb. cosine of the momentum”.

GELFAND-P000967

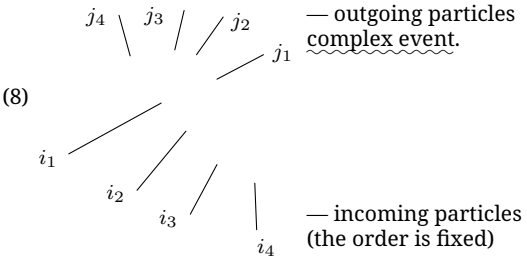
[source number: 20]



Original source details: GELFAND-P000967

Diagram and typed description. In (8) eight separate strokes are drawn, with a gap in the center.

(8)

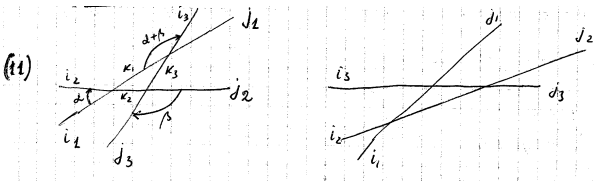


$$S^{j_1 \dots j_4}_{i_1 \dots i_4} = \sum_{\substack{\text{over all possible colorings of internal edges}}} \prod_{\substack{\text{over all vertices}}} S^{\cdot\cdot}_{\cdot\cdot} \tag{9}$$

(10) Lemma.

Postulate B) will always hold if it holds for the triangle.

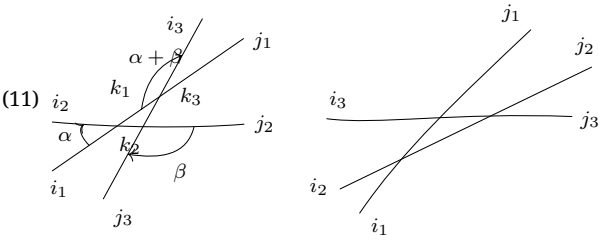
Condition for the $\triangle$.



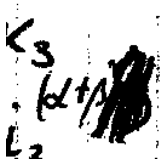
Original source details: GELFAND-P000967

Diagram and typed description. Two arrangements of three lines. The left one labels k_1, k_2, k_3 and the directed arcs $\alpha, \beta, \alpha+\beta$. Only the ends are labelled on the right.

(11)



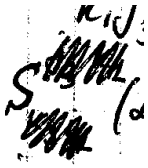
$$S^{k_1k_2}_{i_1i_2}(\alpha)S^{j_1k_3}_{k_1i_3}(\alpha+\beta\,[?])S^{j_2j_3}_{k_2k_3}(\beta) = S^{k_2k_3}_{i_2i_3}(\beta)S^{k_1j_3}_{j_1k_3}(\alpha+\beta)S^{j_1j_2}_{k_1k_2}(\alpha). \tag{12}$$



Original source details: GELFAND-P000967

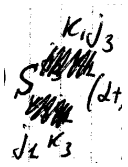
Uncertain reading. After $\alpha+$ part of a sign, probably β, is visible. Heavy cancellation obscures the right side of the argument.

[at the second factor on the right the earlier indices are crossed out. inserted above k_1j_3, below j_1k_3.]



Original source details: GELFAND-P000967

Uncertain reading. The earlier upper and lower indices are crossed out. Individual strokes remain visible, but the full reading is unresolved.



Original source details: GELFAND-P000967

Diagram and typed description. The replacement indices read k_1j_3 above and j_1k_3 below.

Problem: find S, for which these relations hold

The problem is strongly overdet.: eqs. d^6, unknowns d^4.

A simple invar. form of notation:

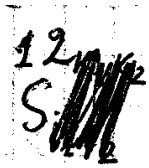
$$A = (a^j_i), \quad B = (b^j_i) \tag{13}$$

i — row, j — column

$$(A \otimes B)^{j\ell}_{ik} = a^j_i b^\ell_k \tag{14}$$

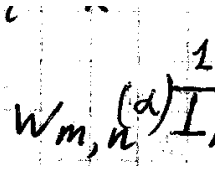
$$\overset{12}{S} = \sum_{m,n=1}^{d^2} w^{(\alpha)}_{m,n} \overset{1}{I}_m \otimes \overset{2}{I}_n \tag{15}$$

[at S the earlier indices are crossed out. 12 is written above S.]



Original source details: GELFAND-P000967

Uncertain reading. The former index pairs are heavily crossed out. Two digits 2 remain visible at the right. The readings j_1j_2 and i_1i_2 are tentative.



Original source details: GELFAND-P000967

Uncertain reading. The parenthesized α sits above m,n. It is set as an upper annotation on $w_{m,n}$.

(16) $\{I_n\}$ — a basis in the space of $d \times d$-matrices M_d.

Indices 1 and 2 indicate the space in which it is taken.

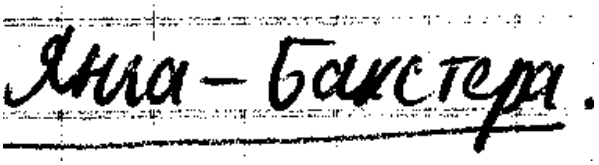
$$\overset{12}{S}(\alpha)\overset{13}{S}(\alpha+\beta)\overset{23}{S}(\beta) = \overset{23}{S}(\beta)\overset{13}{S}(\alpha+\beta)\overset{12}{S}(\alpha) \tag{17}$$

factorization or triangle rel.

(in the tens. prod. of 3 spaces)

(special case of the Yang–Baxter identity)

Yang–Baxter identity: the factorization rel. (i.e. (17)), spoiled in the following ways:



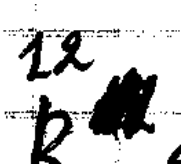
Original source details: GELFAND-P000968

Diagram and typed description. The underlined heading spells the name “Яна-Бакстера”.

$$\overset{121323'}{RSS} = \overset{23'1312}{S\,SR} \tag{18}$$

$$\overset{12}{R} \in \overset{1}{M}_d \otimes \overset{2}{M}_d(\mathbb{C}), \qquad R \in M_d \otimes M_d(\mathbb{C}), \tag{19}$$

[to the upper right of the first R crossed out: entry unresolved; 12 is written above R.]

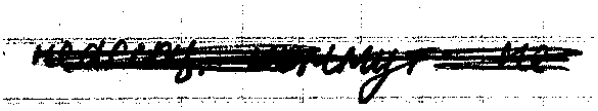


Original source details: GELFAND-P000968

Uncertain reading. The entry to the right of R is crossed out. 12 is clearly written above R.

$$S, S' \in M_d \otimes M_d(D), \tag{20}$$

(D — [crossed out: entry unresolved] an assoc., not necess. commut. alg. over $\mathbb{C}$)



Original source details: GELFAND-P000968

Uncertain reading. Several words before “assoc.” are crossed out. Some letters remain visible, but the phrase is unresolved.

(21) Theorem. From (18) it follows that

(22) [crossed out: entry unresolved]

$$[\mathrm{Sp}_3 \overset{13}{S},\ \mathrm{Sp}_3 \overset{13'}{S}\,] = 0$$



Original source details: GELFAND-P000968

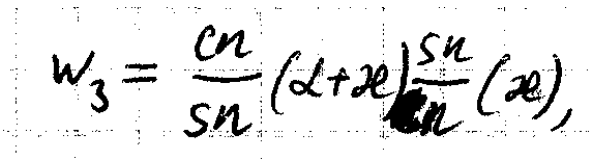
Uncertain reading. The entry before the square bracket is crossed out by diagonal strokes.

(commutator in the space of matrices over the 1st)

Nontriv. example of the factorization rel.:

$$\overset{12}{S}(\alpha) = \sum_{i=0}^{3} w_i(\alpha) \overset{1}{\sigma}{}^i \otimes \overset{2}{\sigma}{}^i \tag{23}$$

$$\begin{aligned} w_1 &= \frac{\operatorname{sn}(x)}{\operatorname{sn}(\alpha+x)}, \qquad w_2 = \frac{\operatorname{dn}}{\operatorname{sn}}(\alpha+x)\frac{\operatorname{sn}}{\operatorname{dn}}(x), \\ w_3 &= \frac{\operatorname{cn}}{\operatorname{sn}}(\alpha+x)\frac{\operatorname{sn}}{\operatorname{cn}\,[?]}(x), \qquad w_0 = 1 \end{aligned} \tag{24}$$



Uncertain reading. The first letter of the denominator in the last fraction for w_3 is heavily overwritten. Tentatively cn, with a possible trace of an earlier dn.

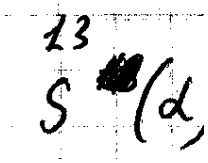
Original source details: GELFAND-P000968

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \qquad \text{— Pauli matr.}$$

$\alpha, x \in \mathbb{C}$.

Instead of (18):

$$\begin{aligned} &\overset{12}{R}(\alpha,\alpha')\overset{13}{S}(\alpha)\overset{23}{S}(\alpha') \\ &\qquad = \overset{23}{S}(\alpha')\overset{13}{S}(\alpha)\overset{12}{R}(\alpha,\alpha') \end{aligned} \tag{25}$$



Uncertain reading. A short entry next to the second S on the right, after the upper 13, is crossed out.

Original source details: GELFAND-P000968

GELFAND-P000969

23/III–81. Gelfand seminar.

I.M. Gelfand on Moser's art.

(coll. in honor of Chern's anniversary)

Ellipsoid

$$\langle A^{-1}x, x\rangle = 1, \qquad \text{semiaxes } \sqrt{\lambda_n} \tag{1}$$

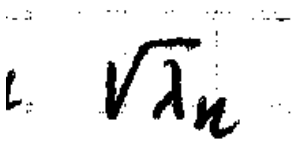


Diagram and typed description. The radical is written with λ_n.

Original source details: GELFAND-P000969

Geodesics:

$$\frac{d^2x}{dt^2} = -\nu A^{-1}x, \qquad x \in \mathbb{R}^n, \tag{2}$$

but for the ellipsoid the point remains on it.

where

$$\nu = \frac{\langle A^{-1}y, y\rangle}{|A^{-1}x|^2}, \qquad y = \frac{dx}{dt}. \tag{3}$$

Isospectral matrix

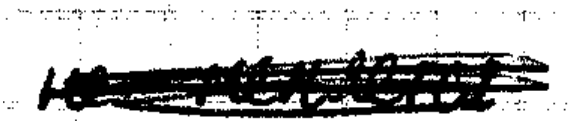
$$L(x,y) = P_y(A - x\otimes x)\cdot P_y \tag{4}$$

P_y — matrix of the orthog. proj. onto the orthog. complement to y:

$$P_y\xi = \xi - \frac{\langle \xi, y\rangle}{|y|^2}y \tag{5}$$

Thm. The matrix

(6) $\underbrace{L\left(x, \frac{dx}{dt}\right)}$ [crossed out: entry unresolved] does not change eigenvalues under motion along a geodesic.

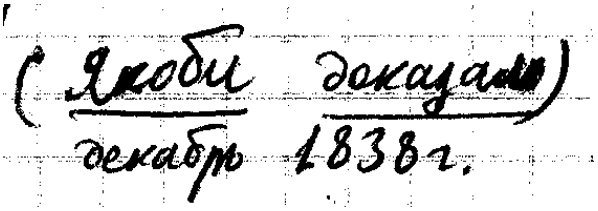


Original source details: GELFAND-P000969

Uncertain reading. The crossed-out entry appears to begin “не” (“not”). The following verb may be “changes” or “depends”, but the strokes obscure the reading.

(Jacobi proved it)

December 1838

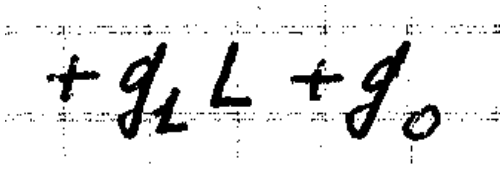


Original source details: GELFAND-P000969

Diagram and typed description. The end of “proved” is retraced, and the date is written separately below.

P. Grinevich.

$$L_6^2 = L_4^3 + g_1L + g_0 \tag{1}$$



Original source details: GELFAND-P000969

Diagram and typed description. The L after g_1 has no subscript.

$$L_4 = \left(\frac{d^2}{dx^2} - u\right)^2 + 2c'_x\big(\wp(\gamma_2) - \wp(\gamma_1)\big)\frac{d}{dx} + \big[c'_x(\wp(\gamma_2) - \wp(\gamma_1)\big]'_x - - \wp(\gamma_1) - \wp(\gamma_2) \tag{2}$$

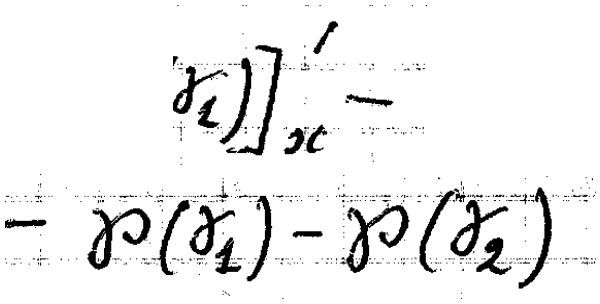
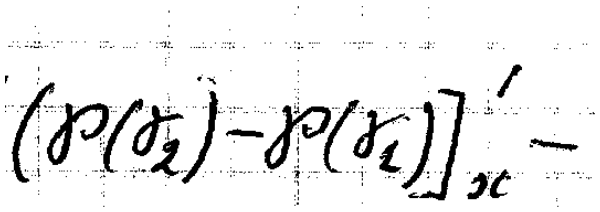


Diagram and typed description. A minus stands at the end of the line and is repeated at the beginning of the continuation.

Original source details: GELFAND-P000969



Uncertain reading. One round parenthesis is visible before the closing square bracket. A separate close for the parenthesis after c'_x is not distinguishable.

Original source details: GELFAND-P000969

$$\begin{cases}\gamma'_1 = \dfrac{1}{\alpha_1 - \alpha_2},\\ \gamma'_2 = \dfrac{1}{\alpha_2 - \alpha_1}\end{cases} \qquad \begin{cases}\alpha'_1 = \alpha_1^2 - u - \varphi(\gamma_1, \gamma_2, 0),\\ \alpha'_2 = \alpha_2^2 - u + \varphi(\gamma_1, \gamma_2, 0)\end{cases} \tag{3}$$

$$\begin{cases}\gamma_1(x) = \gamma_0 + c(x)\\ \gamma_2(x) = \gamma_0 - c(x)\end{cases}$$

$$u(x) = \frac{1}{4{c'_x}^2} - \frac{1}{4}\frac{{c''_{xx}}^2}{{c'_x}^2} - 2\varphi c''_{xx} - {c'_x}^2(\varphi'_c - \varphi^2) + \frac{1}{2}\frac{c'''_{xxx}}{c'_x}$$

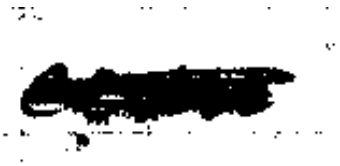
GELFAND-P000970

[source number: 43]

$g_1, g_0, c(x)$ are specified arbitrarily

$$c(x) = \int_{q(x)}^{\infty} \frac{dt}{\sqrt{t^3 + g_1 t + g_0}} \tag{4}$$

$q(x)$ — rat. fn. [crossed out: entry unresolved] $\Rightarrow L_4, L_6$ — oper. with rational coeff.



Uncertain reading. A short entry after “rat. fn.” is crossed out and remains unresolved.

Original source details: GELFAND-P000970

$\gamma_0 = 0 \Rightarrow q(x) = x$. (case of Dixmier)

$$\varphi(\gamma_1, \gamma_2, 0) = \zeta(\gamma_2 - \gamma_1) + \zeta(\gamma_1) - \zeta(\gamma_2)$$

$$\zeta'(z) = -\wp(z), \qquad \zeta(-z) = -\zeta(z).$$

Dixmier's example according to Kirillov

$$L_4 = \{A^2\}, \qquad L_6 = \{A^3\}, \qquad A = L + xL^{-1}, \qquad L = \frac{d^2}{dx^2} + x^3$$

$\{\cdot\}$ — differential part.

[crossed out: ~~Hence~~ (?)] Or:

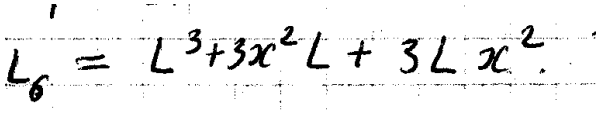


Original source details: GELFAND-P000970

Uncertain reading. The cancelled entry before "Or" may read "Hence".

$$L_4 = L^2 + 2x$$

$$L_6 = L^3 + 3x^2L + 3Lx^2.$$



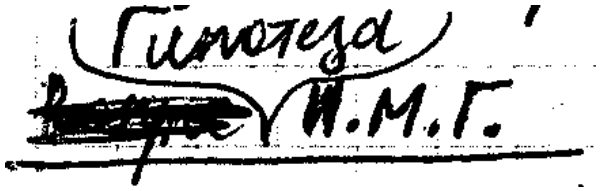
Original source details: GELFAND-P000970

Diagram and typed description. Both terms after L^3 have x^2.

For a curve of genus 2 $P(x, y) = 0$ it is unclear ∃(?) oper. L_1, L_2 with rat. coeff. for which $P(L_1, L_2) = 0$.

There are no coprime ords..

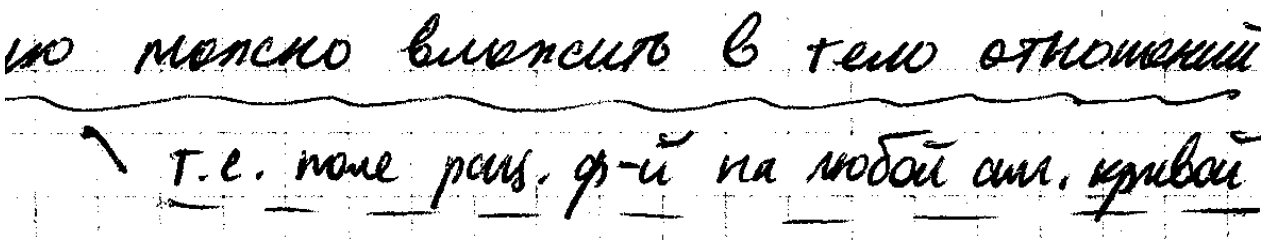
[inserted above the cancelled word: Conjecture] [crossed out: ~~Thm.~~ (?)] I.M.G.: any curve can be embedded in the division ring of fractions of the Weyl alg.



Original source details: GELFAND-P000970

Uncertain reading. "Conjecture" is inserted above the line with a brace. The crossed-out entry before "I.M.G." may read "Thm.".

[marginal addition with arrow] i.e. the field of rat. fns. on any alg. curve

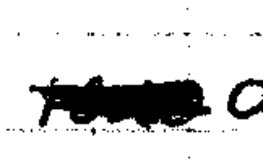


Original source details: GELFAND-P000970

Diagram and typed description. The arrow from the gloss points to the phrase about embedding the curve.

$$q, p, \qquad qp - pq = 1.$$

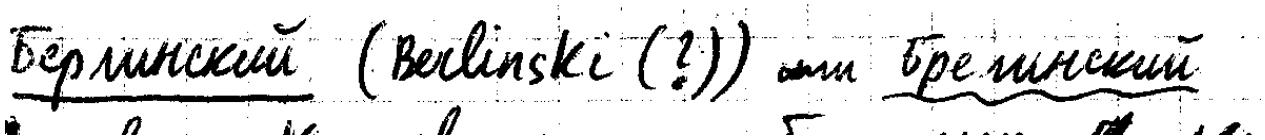
the generated [crossed out: entry unresolved] algebra embeds in a division ring (exercise)



Original source details: GELFAND-P000970

Uncertain reading. A short entry before "algebra" is crossed out.

Berlinskii (Berlinski (?)) or Brelinskii



Original source details: GELFAND-P000970

Diagram and typed description. The manuscript gives two Cyrillic forms of the surname and "Berlinski (?)" in Latin letters.

(Kashiwara's coauthor in the work on the conj. [crossed out: entry unresolved] of Kazhdan–Lusztig)



Original source details: GELFAND-P000970

Uncertain reading. A mark before "Kazhdan" is crossed out. Its reading is unresolved.

$\mathfrak{g}$ — semisimple Lie alg. over $\mathbb{C}$

$\cup$

$\mathfrak{b}$ — Borel alg., $\lambda\colon \mathfrak{b} \to \mathbb{C}$ Lie alg. character

Verma module M_λ (see Dixmier's book on envelop. algebras).

[the title underline is connected to the gloss] geom. objects on the space of flags

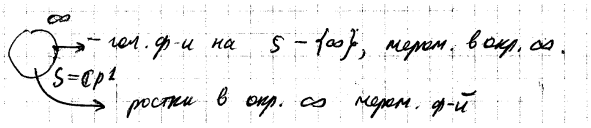
[crossed out: M_λ, $\mathfrak{g} = \mathfrak{b} \oplus \ldots$ (?)]



Original source details: GELFAND-P000970

Uncertain reading. Beneath the strokes one can see M_λ above and tentatively $\mathfrak{g} = \mathfrak{b} \oplus \ldots$ below. The rest is unresolved.

$G = SL(2)$



Original source details: GELFAND-P000970

Diagram and typed description. Circle labelled ∞ with two arrows toward the labels at right.

∞

→ — hol. fns. on $S - \{\infty\}$, merom. near ∞.

$S = \mathbb{CP}^1$

→ germs near ∞ of merom. fns.

GELFAND-P000971

Eckart Viehweg.

(after Hélène Esnault — Exp. Sem. Bourbaki)

(after work by Fujita, Kawamata, Iitaka, Ueno, Viehweg)



Uncertain reading. The initial in Iitaka could be read as I or T. The reading I is preferred.

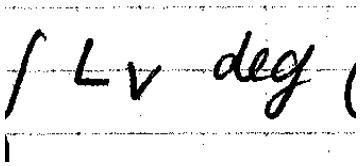
Original source details: GELFAND-P000971

On the classif. of 3-dimensional algebraic varieties.

V — proj. alg. variety over $\mathbb{C}$, $\omega_V = \bigwedge^{\max} \Omega^1_V$

$$K(V) = \begin{cases} L_v \text{ deg} \left(\bigoplus_{n \geq 0} H^0(V, \omega_V^n) \right) - 1 & \text{when } R(V) \neq \mathbb{C} \\ -\infty \end{cases}$$

(Kodaira number



Uncertain reading. The two marks before deg resemble L and a lowered v. The formula suggests the possible reading tr deg.

Original source details: GELFAND-P000971

$$q(V) = \dim H^0(\Omega^1).$$

$$1) \quad \left.\begin{array}{c} V \\ K(V) = 0 \\ q(V) = \dim V \end{array}\right\} \quad (\Leftarrow) \quad V \overset{\text{birat.}}{\sim} \underset{\text{ab. variety}}{A}$$

$$2) \quad \left.\begin{array}{c} \dim V = 3, \\ K(V) = 0 \\ q(V) = \begin{cases} 2 \\ 1 \end{cases} \end{array}\right\} \quad \Rightarrow \quad \alpha : V \longrightarrow \underset{\text{(Albanese variety)}}{\searrow} \quad A(V)$$

étale fibration

$$K(F) = 0.$$

$$3) \quad K(V) = 0, \quad q(V) = 0,$$

$$?$$

4) There is something about the case $K(V) = -\infty$.

Christian Kassel. (Strassburg)

Something about studying the homotopy type of the group Diff(M)

(Waldhausen's idea)

One can extend K-theory to objects A that are rings

$$A \times A \underset{\times}{\overset{+}{\longrightarrow}} A$$

up to homotopy (e.g., loop space)

(distributivity is genuine)

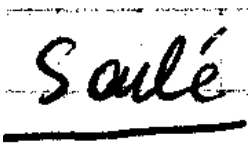
$$\mathbb{Z}[\Omega X]$$

K-functor is constr. using the construction of Quillen.

Reference: AMS Proc. Symp. Pure Math, v. 32 (1976)

Waldhausen, Hatcher.

Christophe Soulé (paris VII)



Uncertain reading. The first vowel in the manuscript surname can be read as o or a. Soulé is the preferred reading.

Original source details: GELFAND-P000971

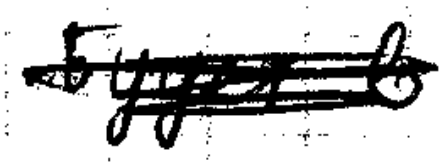
$K_*(\mathbb{Z}) \leftrightarrow \zeta(1-n)$ connection with topoi

GELFAND-P000972

[source number: 44]

$$\#K_{2i-2}(\mathbb{Z}) \geqslant \text{numerator}\,\zeta(1-i)$$
$$\#K_{2i-1}(\mathbb{Z}) \geqslant \text{denominator}\,\zeta(1-i)$$
$$i \equiv 0 \pmod 2$$

[crossed out: ~~Bourb...~~ (?)] Inventiones, 1980, February.



Uncertain reading. The crossed-out writing may begin with “Bourb...”. The rest is unclear.

Original source details: GELFAND-P000972

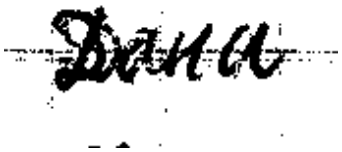
GELFAND-P000973

30/III-81. Gelfand sem.

I. M. Gelfand

Vlasov eq.

Given a mech. system — particles in a potential field.



Uncertain reading. The first letter has been overwritten. The strokes and agreement with “system” favor the reading “Дана” (“given”).

Original source details: GELFAND-P000973

Let there be many particles, and let them interact strongly.

The particles are sparse, so they do not collide, but they create a field — they are all charged. The potential is a function of the particle density.

Arnold. On work by A. Weinstein.

Hydrodynamics:

Consider a region filled with fluid, in a Riem. manifold G.

Motion of the fluid is a curve in the group of diffeom., preserving volume.

$$G = \text{SDiff}(G)$$

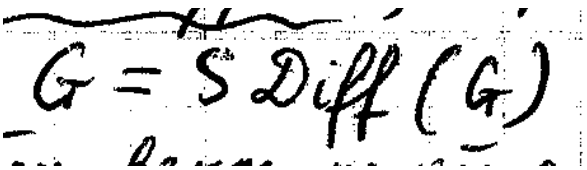


Diagram and typed description. Both letters have the form G. There is a short stroke beneath the letter in parentheses.

Original source details: GELFAND-P000973

$\mathfrak{g}$ — Lie alg. of G, i.e. the algebra of vector fields with divergence $= 0$.

Energy: $E = \frac{1}{2}\int |v(x)|^2\,dx$, $v(x)$ — is a vector field.

This is a quadr. form on $\mathfrak{g}$.

Extend it by left translations to all of G. We get a Riem. metric

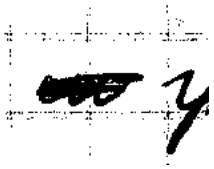
Def. Geodesics are flows of an ideal fluid

(Arnold, 1966)

Ann. Inst. Fourier and a survey in the book on mechanics.

There is curvature, a Hamiltonian formalism, etc.

Weinstein — extension to other [crossed out: writing unclear] groups (∞-dimensional)



Uncertain reading. The writing beneath the heavy cancellation before “groups” remains unclear.

Original source details: GELFAND-P000973

Symplectic group: gives the Vlasov eq.

(cold plasma)

Extend, add local degrees of freedom — get various eqs.

(including Yang–Mills).

if we take a gauge group.

Hamiltonian:

1) On T^*G there is the standard sympl. structure;

E defines the Hamiltonian.

$\Rightarrow$ Euler eq.

GELFAND-P000974

[source number: 59]

2) (2nd version)

Consider the map

$$T^*G \longrightarrow \mathfrak{g}^*$$

(one set of translations, left or right)

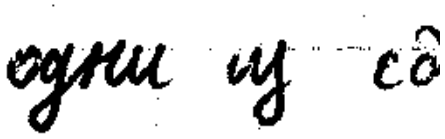


Diagram and typed description. The first word reads “одни” (“one set”).

Original source details: GELFAND-P000974

The curves from 1) project to the Euler eq. in $\mathfrak{g}^*$.

Symplectic struct. on the orbits of the coadjoint representation

(Kirillov struct.).

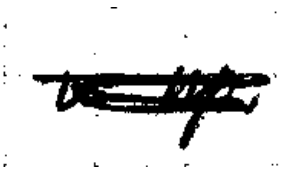
The Hamiltonian transfers by invariance.

The Kirillov form descends.

For a compressible fluid: Ebin and Marsden (later)

rot. is given $\Rightarrow$ only then do we get a Hamiltonian formalism

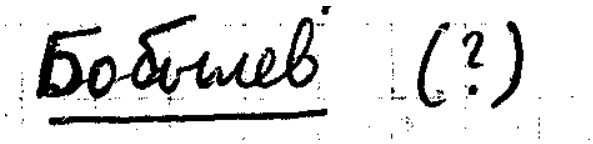
over G one must take bundles [crossed out: writing unclear] (where the density is)



Original source details: GELFAND-P000974

Undeciphered passage. The writing after "bundles" is heavily crossed out.

Someone named Bobylev (?)

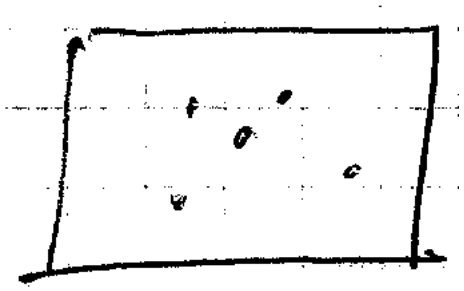


Original source details: GELFAND-P000974

Diagram and typed description. The parenthesized question mark is written beside the surname.

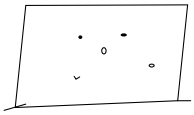
On the Boltzmann eq.

Gas in a vessel.



Original source details: GELFAND-P000974

Diagram and typed description. Vessel with five particle marks.



The spheres (hard) are the interaction potential $V(r)$.



Original source details: GELFAND-P000974

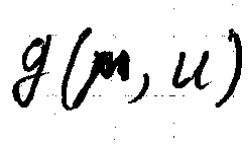
Uncertain reading. A short vertical stroke stands above the end of "interaction". Its meaning is unclear.

$f(\vec{v}, t)$ — the number of particles with velocity $\vec{v}$ per unit volume at the given instant.

$$\frac{\partial f}{\partial t} = \int d\vec{w}\, d\vec{n}\, g\left(\frac{\vec{u}\vec{n}}{u}, u\right)\{f(\vec{v}')f(\vec{w}') - f(\vec{v})f(\vec{w})\} \tag{1}$$

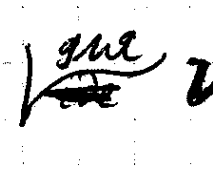
$$\begin{aligned} |\vec{n}| = 1, \qquad & \vec{u} = \vec{v} - \vec{w}, \\ \vec{v}' = \tfrac{1}{2}(\vec{v} + \vec{w} + u\vec{n}), \qquad & \vec{w}' = \tfrac{1}{2}(\vec{v} + \vec{w} - u\vec{n}) \end{aligned} \tag{2}$$

$$g(\mu, u) = \widetilde{g}(\mu) u^{1-\frac{4}{s}}, \quad \text{for} \quad V(r) = \frac{\alpha}{r^s} \tag{3}$$



Original source details: GELFAND-P000974

Uncertain reading. The first argument of g is heavily retraced. The adjacent $\widetilde{g}(\mu)$ favors μ.



Original source details: GELFAND-P000974

Uncertain reading. “For” was inserted above a crossed-out word that may read “where”.

(this is for the spatially homog. case)

In fact

$$g = u\frac{d\sigma}{d\Omega}, \qquad \sigma \text{ — cross section} \tag{4}$$

$(\vec{v}', \vec{w}')$ — velocities after the collision

$(\vec{v}, \vec{w})$ — —‖— before —‖—

$\vec{u}$ — relative velocity, it turns and gets a new dir. $u\vec{n}$ (magnitude is conserved by conservation of energy)

Quadr. form — because the probability of collision is proportional

[crossed out: writing unclear] In recent years there has been much progress: if g depends only on the angle



Original source details: GELFAND-P000974

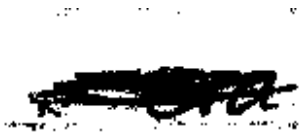
Undeciphered passage. The short entry before “In recent years” is heavily crossed out.

(i.e. $V(r) = \frac{1}{r^4}$) $\Rightarrow$ greatly simplified using the Fourier transform. Then

we get exact solutions, an inf. number of conservation laws, etc.
Convolution with a Gaussian density does not change the eq.
Eigenfns. of the linearized eq. were obtained (new ones).
Consider

$$f_M = A \exp[-\alpha(\vec{v} - \vec{v}_0)^2] \tag{5}$$

[crossed out: writing unclear] — stationary



Undeciphered passage. The short entry between (5) and the dash is heavily crossed out.

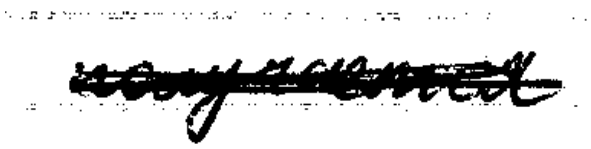
Original source details: GELFAND-P000975

— a 3-param. family of fns., for the parameters $A, \alpha, \vec{v}_0$ [crossed out: writing unclear]



Diagram and typed description. The manuscript reads “3-param.” before “family”.

Original source details: GELFAND-P000975



Undeciphered passage. The entry after the list of parameters is heavily crossed out. The letters beneath the stroke are unclear.

Original source details: GELFAND-P000975

$$f = f_M[1 + F], \qquad F = F(\vec{v}, t), \qquad \|F\| \ll 1. \tag{6}$$

Hence for F we get the linearized eq.

$$\frac{\partial F}{\partial t} = \widehat{L}F, \tag{7}$$

where

$$\widehat{L}F = \int d\vec{w}\, e^{-w^2/2}\, d\vec{n}\, g\left(\frac{\vec{u}\vec{n}}{u}, u\right)[F(\vec{v}') + F(\vec{w}') - F(\vec{v}) - F(\vec{w})] \tag{8}$$

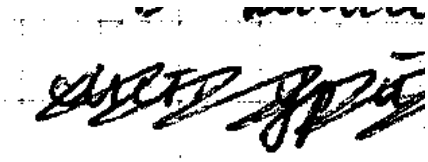
The general solution is constructed in the class of rapidly decaying fns. [crossed out: ~~An infinite~~ (?) writing unclear]



Undeciphered passage. “An infinite” is a possible reading in the crossed-out entry after “the general solution”. The whole entry remains unclear.

Original source details: GELFAND-P000975

[crossed out: ~~system of eqs.~~ (?) writing unclear]



Original source details: GELFAND-P000975

Uncertain reading. The start of the crossed-out line tentatively reads “system of eqs.” The ending of the second abbreviation is unclear.



Original source details: GELFAND-P000975

Undeciphered passage. The rest of the crossed-out line remains unclear.

(9) $s = 4$; [crossed out: writing unclear] $f(v^2)$ — isotropic case $\Rightarrow$ reduces to an ∞ system of ordinary eqs. A transform takes it to a linear system.



Original source details: GELFAND-P000975

Uncertain reading. The separate entry between $s = 4$ and $f(v^2)$ is heavily crossed out.

Unknown in the other cases.

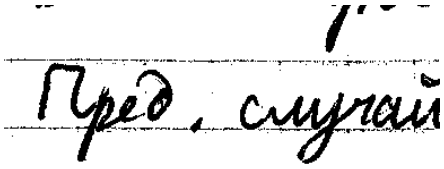
This is analogous to eqs. with const. coeffs. — here g does not depend on u.

$$\widehat{L} = CI + T, \qquad T \text{ — a compact operator.} \tag{10}$$

In this case $C = -1$.

The spatially inhomog. case is important.

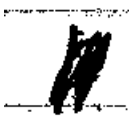
Lim. case (in mean free path) — Navier–Stokes eq.



Original source details: GELFAND-P000975

Uncertain reading. The manuscript has the abbreviation “Пред.”. It is taken here to mean “limiting”.

B. Lidskii. [crossed out: writing unclear]



Original source details: GELFAND-P000975

Uncertain reading. A short mark to the right of the heading is crossed out. Its reading remains unclear.

Consider [crossed out: $\mathbb{R}^{3n}$ (?)] $(x_1, \dots, x_n, y_1, \dots, y_n, z_1, \dots, z_n)$,



Original source details: GELFAND-P000975

Uncertain reading. The notation crossed out before the coordinates may be $\mathbb{R}^{3n}$.

$$\sum_{i=1}^{n}(x_i + y_i - z_i) = 0 \tag{1}$$

Consider there a subset singled out by inequalities

$$x_1 \geqslant \ldots \geqslant x_n, \quad y_1 \geqslant \ldots \geqslant y_n, \quad z_1 \geqslant \ldots \geqslant z_n \tag{2}$$

and $\exists$ 3 Hermitian matrices X, Y, Z, for which

$$X + Y - Z = 0 \tag{3}$$

and their spectra are subsets of (2).

GELFAND-P000976

[source number: 60]

(4) Definition: Let a set be given
(by induction)

$$I_n = \{1, 2, \ldots, n\} \qquad I_\ell = \{1, 2, \ldots, \ell\}$$

and 3 increasing seqs. of its elems.

$$\begin{gathered} a_1 < a_2 < \ldots < a_\ell \\ b_1 < b_2 < \ldots < b_\ell \\ c_1 < c_2 < \ldots < c_\ell \\ (a_i, b_i, c_i \in I_n) \end{gathered}$$

They are consistent if

$$1) \qquad \sum_{i=1}^{\ell}(a_i + b_i - c_i) = \frac{\ell(\ell+1)}{2}$$

2) Suppose 3 consistent seqs. are given

$$\begin{gathered} f_1 < f_2 < \ldots < f_k \\ g_1 < g_2 < \ldots < g_k \\ h_1 < h_2 < \ldots < h_k \end{gathered} \qquad \begin{gathered} (\text{of elements of } I_\ell). \\ \ell < n \end{gathered}$$

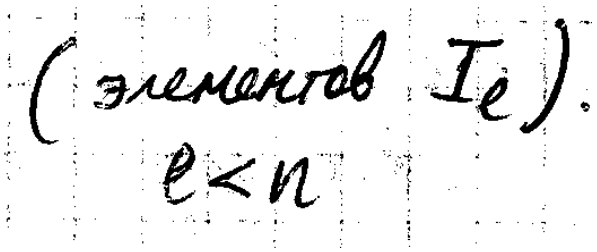


Diagram and typed description. Below "of elements of I_ℓ" the manuscript has $\ell < n$.

Original source details: GELFAND-P000976

and they are consistent. Then we must have

$$\sum_{j=1}^{k} a_{f_j} + \sum_{j=1}^{k} b_{g_j} \leqslant \sum_{j=1}^{k} c_{h_j} + \frac{k(k+1)}{2}$$

— · —

Final answer:
The inequalities that must hold are (2) and

$$\sum z_{c_j} \leqslant \sum x_{a_j} + \sum y_{b_j} \tag{5}$$

— · —

$a_i \mapsto (a_i - i) \Rightarrow$ then everything is written more simply.
Probability is also written as a convolution.

— · —

There is a dual theorem ⇒ an algorithm to determine which reps. occur in the tensor prod.



Uncertain reading. The abbreviation before “in the tensor” could be “встрет.” or “встреч.”, both indicating occurrence.

Original source details: GELFAND-P000976

GELFAND-P000977

6/IV-81. Gelfand sem.
S. G. Krein. On the eqs. of a compressible fluid.
(someone’s work) — Ebin, Marsden

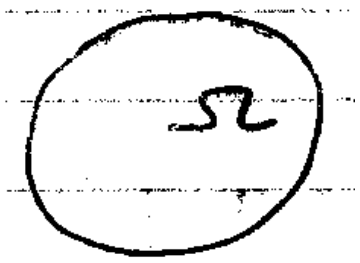
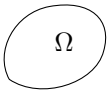


Diagram and typed description. Closed outline labelled Ω.



Original source details: GELFAND-P000977

diffeomorphisms of the region onto itself arise G.

Incompress. — preserve volume — group G_0.

G_0 — Riem. manifold, motion along geodesics.

Barotropic fluid: $p = p(\rho)$, e.g. $p = A\rho^\gamma$.

T is taken $=$ const.

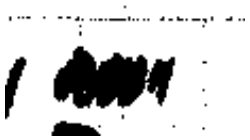


Diagram and typed description. The line about taking T constant is written separately.

Original source details: GELFAND-P000977

Eqs.

$$\begin{cases} \dfrac{\partial \vec{u}}{\partial t} + (\vec{u}, \nabla)\vec{u} = -\dfrac{1}{\rho}\operatorname{grad} p \\ \dfrac{\partial \rho}{\partial t} + \operatorname{div}(\rho \vec{u}) = 0 \\ p = p(\rho) \end{cases} \tag{1}$$



Undeciphered passage. There are heavily crossed-out marks above p in the first line. Their reading remains unclear.

Original source details: GELFAND-P000977

I.M.G. thinks that barotropy is a poor restriction.

(heating, of a pump while pumping)

Next: Shock waves necessarily arise and must be taken into account.

$$u_t + cu_x = 0 \quad \Rightarrow \quad u = f(x - ct)$$



Diagram and typed description. A pulse with an almost flat top. A rightward arrow is labelled c.



Original source details: GELFAND-P000977

$$u_t + uu_x = 0 \quad \Rightarrow \quad u = f(x - ut)$$

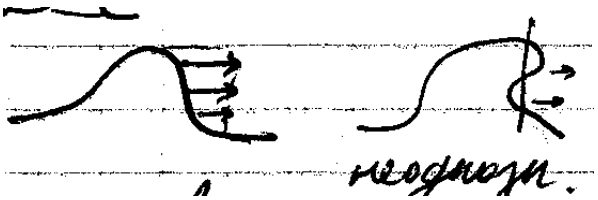
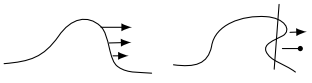


Diagram and typed description. At left, three rightward arrows lie on the falling side. At right, the curve folds around a vertical stroke. The upper mark to its right has a rightward arrowhead. The lower mark thickens at the right and its arrowhead is unclear.



multival.

Original source details: GELFAND-P000977

Riemann. At this point the soln. becomes discontinuous. We must write

$$\frac{\partial u}{\partial t} + \frac{1}{2}\frac{\partial (u^2)}{\partial x} = 0$$

and understand what a discontinuous solution is

GELFAND-P000978

[source number: 68]

Zakharov: eq.

$$\underbrace{\frac{\partial U}{\partial t} + U\frac{\partial U}{\partial x} = 0}$$

where U — a matrix,

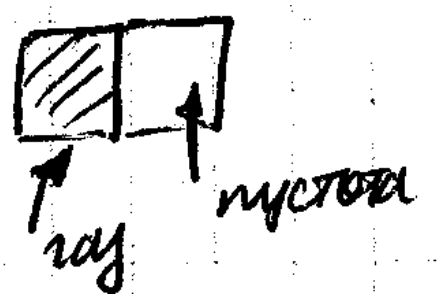
can also be integrated completely.

[crossed out at left: ~~Ex~~ (?)] Discontinuous solutions are always needed in nonlinear hyperb. eqs.



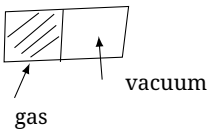
Original source details: GELFAND-P000978

Uncertain reading. The crossed-out mark at left may read “Пр” (“Ex”).



Original source details: GELFAND-P000978

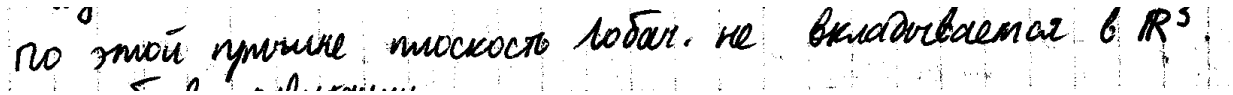
Diagram and typed description. A partition divides the vessel. The hatched left side is marked “gas” and the right side “vacuum”.



Remove the partition. How does it even out?

In fact, a system of shock waves going back and forth.

For this reason the Lobachevsky plane does not embed in $\mathbb{R}^3$.



Original source details: GELFAND-P000978

Diagram and typed description. The manuscript reads “does not embed”.

And esp. in gravitation.

S.G.K. weakly compress. fluid (there is a parameter.)

In the limit the solutions reduce to solns. of the eqs. of an incompress. fluid.

Hamiltonian system. Density is close to 1.

$g(x)$ — Jacobian of the map

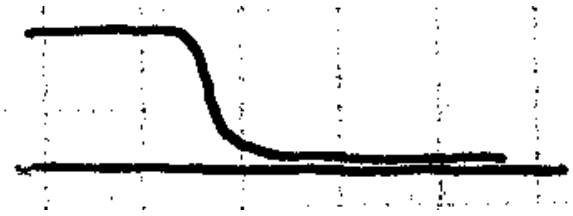
$$\int_\Omega \int_1^{g^{-1}(x)} \frac{p(\rho)}{\rho^2}\, d\rho\, dx \qquad \text{— potential energy}$$

The same exists for a viscous fluid (following Arnold).

Problem (I.M.G.).

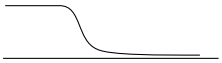
$$u_t + uu_x + p\left(\frac{\partial}{\partial x}\right) u = 0$$

$u|_{t=0} =$ [graph]



Original source details: GELFAND-P000978

Diagram and typed description. The curve drops from an upper horizontal stretch to a lower one, above a separate horizontal line.



What will happen?

KdV — solitons keep splitting off.

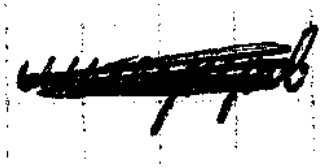
G. Olshanskii. Lie semigroups.

[crossed out: ~~Everywhere~~ (?)] This is useful for [crossed out: writing unclear] the theory of reps. of ∞-dimensional groups.



Original source details: GELFAND-P000978

Undeciphered passage. The crossed-out word before “This” may read “Everywhere”.



Original source details: GELFAND-P000978

Undeciphered passage. The entry between “for” and “theory” is heavily crossed out.

$\mathfrak{g}$ — semisimple Lie algebra.

Example of a Lie semigroup.

$\mathbb{C}^n$, $n = p + q$, consider an indefinite scalar product

$$\langle x, x\rangle = -|x_1|^2 - \cdots - |x_p|^2 + |x_{p+1}|^2 + \cdots + |x_n|^2$$

$A\colon \mathbb{C}^n \to \mathbb{C}^n$ is called a J-contraction, if

$$\langle Ax, Ax\rangle \leqslant \langle x, x\rangle, \quad x \in \mathbb{C}^n.$$

This is a semigroup (working example).

Real example.

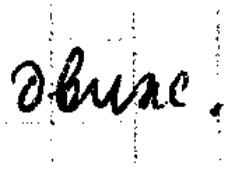


Original source details: GELFAND-P000978

Diagram and typed description. A slanting boundary with seven strokes on the right.

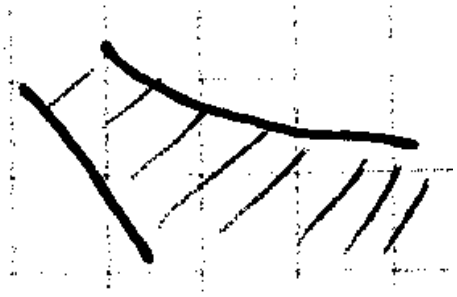


consider motions of the Lobachevsky plane that send into itself some half-plane



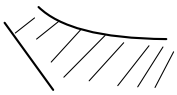
Original source details: GELFAND-P000978

Diagram and typed description. The manuscript abbreviates the word for “motions”.



Original source details: GELFAND-P000978

Diagram and typed description. A slanting line at left and a curve above. Slanted hatching lies between them and below the curve.



Theorems

1) description of Lie semigroups.

2) description of all reps. of complex $\mathfrak{g}$.

3) —||— real forms.

4) D — bounded symm. domain, e.g., the ball $|z| < 1$.

$D \subset \widehat{D}$. (dual variety)

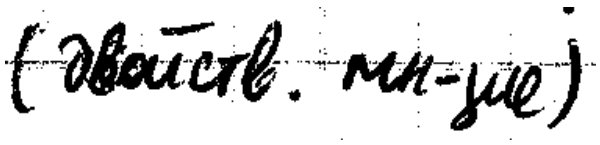


Diagram and typed description. The parenthesis reads “dual variety” in abbreviated form.

Original source details: GELFAND-P000979

G — semigr. of transformations of $\widehat{D}$, taking D into the domain

$z \in \mathrm{Mat}_{p,q} \Rightarrow$ a semigr. arises of J-contractions

— · —

Γ — semigr. of matrices A with $\|A\| \leqslant 1$.

$x \mapsto x^{\#}$ involution

$$(ab)^{\#} = b^{\#}a^{\#}, \qquad a^{\#\#} = a.$$

(here it is $*$).

A semigroup rep. is a homomorphism

$$T\colon x \mapsto T(x) \quad \text{into the set of contractions of a complex Hilbert space,}$$

$$T(x^{\#}) = T(x)^* \quad \text{— analogue of unitarity}$$

$$\|T(x)\| \leqslant 1.$$

$$T(xy) = T(x)T(y)$$

Semigr. of contractions $\mathbb{C}^n = V$.

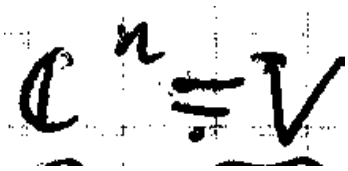


Diagram and typed description. A distinct dot lies below the equality sign.

Original source details: GELFAND-P000979

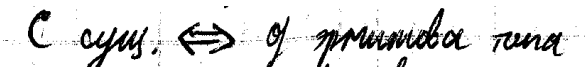
Reps. in $V^{\otimes m}$ are indexed by sequences — these are analytic.
There are also antianalytic ones.
More interesting are J-contractions.
Here analytic reps. arise.

— · —

$\mathfrak{g}$ — semisimple Lie algebra;

$$\underset{\substack{\text{cone in } \mathfrak{g} \\ \text{(convex)} \\ \text{with nonempty interior}}}{C} \subset \mathfrak{g}$$

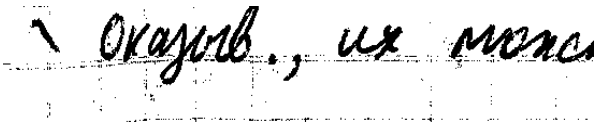
Invar. — invariant under the adjoint representation.
$\exists$ Do such cones exist? indep. Vinberg and Kostant.
Let $\mathfrak{g}$ be simple. C exists $\Leftrightarrow$ $\mathfrak{g}$ of Hermitian type



Original source details: GELFAND-P000979

Diagram and typed description. The two conditions are joined by an equivalence sign.

(the corresp. Riem. symm. space has a complex structure)
Vinberg found max and min.
[stroke leading to the remark] It turns out, they can be described!!

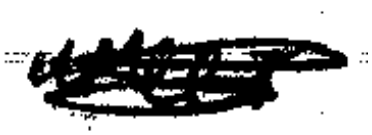


Original source details: GELFAND-P000979

Diagram and typed description. The remark starts with an abbreviation for “It turns out”.

$$\mathfrak{g} = \underset{\text{compact}}{k} + p$$

[crossed out at right: writing unclear]



Original source details: GELFAND-P000979

Undeciphered passage. The entry to the right of “compact” is heavily crossed out.

Cond. k has a 1-dimensional center.
their list:
1) $\mathrm{su}(p,q)$
2) $\mathrm{sp}(2n,\mathbb{R})$
3) $\mathrm{so}^*(2n)$
4) $\mathrm{so}(2,n)$
and 2 exceptional ones.

GELFAND-P000980

[source number: 69]

Consider the complexification $G_{\mathbb{C}}$ and the subset

$$G \cdot \exp iC \subset G_{\mathbb{C}}.$$



Uncertain reading. A subscript of the first G is crossed out. Its reading remains unclear.

Original source details: GELFAND-P000980

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1) Kostant–Vinberg theorem.

[crossed out at left: writing unclear] a cone C (conv. invar.) in a simple real Lie alg. $\mathfrak{g}$ with nonempty interior

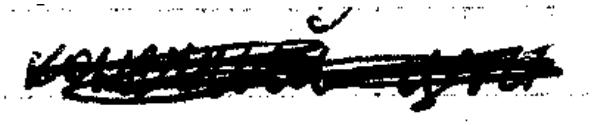
exists $\Leftrightarrow$



Uncertain reading. A mark to the left of the line about the cone is crossed out.

Original source details: GELFAND-P000980

$\mathfrak{g}$ is of Hermitian type (i.e. the max. compact subalgebra has a [crossed out: writing unclear] 1-dimensional center)



Undeciphered passage. Several strokes are visible under the cancellation before "1-dimensional center". Their reading remains unclear.

Original source details: GELFAND-P000980

[crossed out at left: writing unclear] I.M.G. It is wrong to restrict to simple and semisimple ones (in ∞-dimensional theory)



Uncertain reading. A mark to the left of "I.M.G." is crossed out. Its reading remains unclear.

Original source details: GELFAND-P000980

2) Among all cones there are $C_{\max}$ and $C_{\min}$, $C_{\max} \supset C_{\min}$

(unique)



Uncertain reading. The digit 2 is retraced over other strokes.

Original source details: GELFAND-P000980

(G. Olshanskii and simult. S. M. Paneitz — a student of I. Sigal.)

There is a description ↗ [arrow to the preceding line] in terms related to the Cartan subalgebra.

3) Consider the set

$$G \exp iC \subset G_{\mathbb{C}}.$$

[crossed out at left: writing unclear] Thm. $\forall$ cones C the set $G \exp iC$ — a semigroup!



Original source details: GELFAND-P000980

Uncertain reading. A short entry to the left of "Thm." is heavily crossed out.

A weaker version holds for $\forall$ Lie algebras. We get a local Lie semigr.

$$\begin{array}{l} G \exp iC \\ \| \\ \Gamma(C) \end{array} \quad \text{— the class of complex semigroups considered}$$

$$\Gamma(C) \supset \underset{\text{(interior)}}{\Gamma^{\circ}(C)}$$

A holom. rep. if the matrix elems. are holom. in the interior.

Irred. = tensor prod. of holom. and antiholom..

[written below the abbreviation: irred. rep.]

Holom.: distinguish (?) a highest-weight rep..

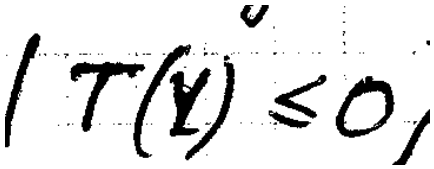


Original source details: GELFAND-P000980

Uncertain reading. The last letter after "выде" is heavily retraced. "выдел." and "выдер." are possible readings.

G — group; T — unit. representation, $t(x)$, $x \in \mathfrak{g}$.

$$C(T) = \{Y \mid T(Y) \leqslant 0\} \qquad t(y), \quad y \in i\mathfrak{g}.$$



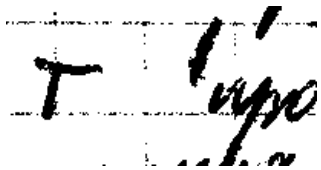
Original source details: GELFAND-P000980

Diagram and typed description. The cone formula shows an uppercase T. The nearby letters with shorter crossbars are rendered as t.

a highest-weight rep. $\Leftrightarrow C(T)$ is nontrivial.

Thm. T — a highest-weight rep., $C = C(T)$ — corresp. cone.

Then T 'extends to a holom. rep. of the corresp. semigroup $\Gamma(C)$

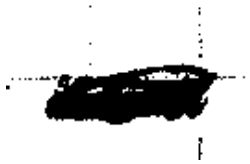


Original source details: GELFAND-P000980

Uncertain reading. There is a raised stroke before “extends”. Its meaning is unclear.

This is the exact domain of extendibility.

category of holom. reps. [crossed out: writing unclear] of the semigroup $\overset{\text{isom.}}{\Leftrightarrow}$ category of highest-weight reps. of the group.



Original source details: GELFAND-P000980

Undeciphered passage. The writing before “of the semigroup” is covered by crossing out.

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